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International Journal of Mobile and Blended Learning

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MISSION

The primary mission of the **International Journal of Mobile and Blended Learning (IJMBL)** is to provide insight and understanding into the role of innovative learning theory and practice in an increasingly mobile and pervasive technological environment. As technology enables a more seamless experience of device-supported learning worlds that may integrate mobile, embedded, augmented, and immersive technologies, researchers, professionals, and academicians may expect to see increasing interest and activity in blended approaches to learning. IJMBL brings together experts at the forefront of this field, in both technology and pedagogical practice, and assists them in the development and dissemination of new approaches to both mobile and blended learning.

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Editorial Preface

David Parsons, The Mind Lab, Auckland, New Zealand
Kathryn MacCallum, University of Canterbury, Christchurch, New Zealand

Welcome to volume 13 issue 3 (Q3 2021) of the International Journal of Mobile and Blended Learning. The five articles in this issue combine the work of authors from 5 predominantly Asian countries and cover such diverse topics as collaboration. reading, stress, learning styles and gamification.

Our first article in this issue is "Development Trends and Analysis of Collaborative Learning in e-Learning Environments" by Chun Chao Shih (National Changhua University of Education, Taiwan) and Ying Chih Kuo (Hsing Wu University, New Taipei City, Taiwan). This bibliometric literature review examines trends in collaborative e-learning research. The analysis uncovers research trends in topic-related sequence and geographic differences in sub-topics by space, with further analysis on the structure of knowledge bases. Research topics are revealed to be cross-border, not limited to geographically close nations.

The second article is "Influencing Factors of Intention to Use Mobile Devices for Reading: Moderation Effects of Perceived Sociality and Perceived Risk" by Pinghao Ye and Liqiong Liu of the Wuhan Business University, China. This research used the stimulus–organism–response model as basis to build a causal model of mobile readers and their continuous intention to use this mode of reading. Perceived pleasure, perceived service, immersion and perceived sociality were revealed to have a significant positive effect on intention for continuous use. Several other positive and regulating effects were also explored in the data analysis.

Article 3 is "The Dark Side of Technology-Enabled Teaching: Impact of Technostress on Student Performance" by Sangeeta Mehrolia, Subburaj Alagarsamy and Jeevananda S, all from Christ University, Bangalore, India. This article investigates the impact of technostress on students. The Stimulus, Organism and Response Model is used to analyse the mediating effect of technostress creators and understand how technostress inhibitors influence students' satisfaction and their performance. It is hoped that insights from this study will enable higher education institutions to identify any students who are finding technology-based education problematic and help them to reduce stress in order to enhance their active participation in learning.

Our fourth article is "Blended Learning in Teaching Technical Teacher Candidates With Various Types of Learning Styles" by Hamonangan Tambunan, Marsangkap Silitonga and Uli Basa Sidabutar, all from the State University of Medan, Indonesia. Although the topic of learning styles is somewhat controversial, the authors of this study note that there are various aspects of learning styles that are still valid to considered. The authors compare the different impacts of the balance of face to face and online learning in blended learning, along with four learning style type groups of Diverger, Assimilator, Converger, and Accommodator. It was found that both the blend of online and face to face learning and the types of learning styles can significantly impact competence outcomes.

The fifth and final article is "Gamified E-Reading Experiences and Their Impact on Reading Comprehension and Attitude in EFL Classes" by Aysegul Liman Kaban of Bahcesehir University, Istanbul, Turkey. This article looks at the potential benefits of using gamified e-reading tools with secondary school EFL learners. It reports on a study that measured the perceptions and reading comprehension performance of participants in different conditions. The findings revealed that while there were positive outcomes in class, the students still preferred printed books rather than electronic books for leisure, suggesting that the affordances of both reading media have advantages in different contexts.

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This issue of the journal underlines that mobile and blended learning are truly global research areas, and we particularly welcome articles the explore that global experience as well as the specific local conditions that can impact on teaching and learning with technology. We look forward to receiving similarly diverse submissions for future issues of IJMBL.

David Parsons
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International Journal of Mobile and Blended Learning
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Development Trends and Analysis of Collaborative Learning in E-Learning Environments 1988-2019

Chun Chao Shih, Department of Industrial Education and Technology, National Changhua University of Education, Taiwan
Ying Chih Kuo, Hsing Wu University, Taiwan

ABSTRACT

This study applies the bibliometric method to review research in collaborative learning in e-learning and analyzes the trends of research on this topic. Using quantitative tools of science mapping, 8,575 papers in the Scopus database, prior to and including 2019, were reviewed, tracing back to 1988. Retrospective analysis uncovers continuing trends in research by way of topic-related sequence and geographic differences in sub-topics by space; moreover, further analysis is undertaken on the structure of knowledge bases. This reveals that the journals of highest impact include Computers in Human Behavior*,* Computers & Education*, and* Journal of Computer Assisted Learning*, while the most impactful authors are Barolli, Caballé, Chen, Daradoumis, Dimitriadis, Li, Li, Sterbini, Temperini, Tsiatsos, and Xhafa. This paper concludes that the bibliometric method can target a broad range of research; topics related to applied science and emerging technologies are still to be studied. Research topics are cross-border, not limited to geographically close nations.*

KEYWORDS

Bibliometric Method, Collaborative Learning, E-Learning

1. INTRODUCTION

Most past review papers of research on Collaborative Learning in e-Learning have included just a few countries in their sample (Al-Samarraie & Saeed, 2018; Chavez & Romero, 2012; Dado & Bodemer, 2017; Jeong, Hmelo-Silver & Yu, 2014; Schwendimann, De Wever, Hämäläinen, & Cattaneo, 2018). In view of the massive differences in societal transition over time and the widely differing circumstances in the research world of each country, it is not a given that research results are highly mobile, no matter in terms of time or space. Previously, Tang, Tsai and Lin (2014) conducted a co-citation network analysis on 2006-2013 research publications on topics related to computer-supported collaborative learning, exploring relevant sequences; however, no more comprehensive evaluation has been carried out in this field. Thus, this paper employs a search for synonymous keywords frequently used in recent years, expands the time span of the searches to conduct a comprehensive evaluation through time Sequence, geographic Space and knowledge base Structure. Hence, through the application of scientific landscape mapping, the knowledge base related to Collaborative Learning in e-Learning are analyzed to answer the following research questions:

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Question 1: What is the status of research literature volume and its growth in the area of Collaborative Learning in e-Learning? [corresponding to Sequence]
Question 2: What are the geographic origins of research in the area of Collaborative Learning in e-Learning according to volume and distribution? [corresponding to Space]
Question 3: Which journals, authors, and papers have the biggest impact on research in the area of Collaborative Learning in e-Learning? [corresponding to Structure]

The terminology in research topics related to Collaborative Learning in e-Learning in recent years is not exactly the same across the board; for example: Collaborative e-learning (Calle-Alonso et al., 2017; Marie O. Robles, 2017; Salim & Systems, 2017), Computer-supported collaborative learning (Chen, Wang, Kirschner, & Tsai, 2018; Jeong, Hmelo-Silver, & Jo, 2019; Xia & Borge, 2019), Online collaborative learning (Fernández-Ferrer & Cano, 2016; Muuro, Oboko, & Wagacha, 2016; Saqr, Fors, & Tedre, 2018), and so forth, all include the meaning of learners learning together with the aid of digital tools. In this paper, e-CL is used as an abbreviation of this theme. Quantitative tools of scientific landscape mapping (Small, 1999) are used in this paper to search e-CL papers in the Scopus database, prior to and including 2019; and the bibliometric method (Zupic & Čater, 2015) is used for analysis. This study aims to analyze and provide understanding about the global e-CL knowledge base.

2. CONCEPTUAL FRAMEWORK

First, the dimension of Sequence is analyzed on papers already published, through which we can see the growth or diminishing of related research, which can reveal the research publication growth trajectory. (Hallinger, 2018b). Second the Space dimension of already published research results is analyzed, which refers to the geographic distribution status. This is important as current research review papers are mostly focused on results from research in a minority of countries (Al-Samarraie & Saeed, 2018; Chavez & Romero, 2012; Dado & Bodemer, 2017; Jeong et al., 2014; Schwendimann et al., 2018) rather than being representative of all nations around the world. However, there are researchers from more and more countries around the world actively conducting e-CL research, so it is obviously of increasing importance to understand the pulse of related research in diverse nations. Thirdly, the Structure dimension is analyzed, that is the intellectual structure of the subjects comprising this field of research, research topics and the relationship between them (Zupic & Čater, 2015). Researchers in this study make use of research paper sources, authors, titles and keywords in the analysis.

Employing what the authors term as the 3S conceptual framework of Sequence, Space and Structure (Figure 1) described above, this research conducts an analysis of the e-CL knowledge base to answer the three research questions posed in this paper with the purpose of discovering the locus of the knowledge base as well as its scope and boundaries. Formation of the Sequence dimension can reveal trends of past and future research, while the Space dimension can illustrate the research status and differences of each country, whereas which journals, authors and papers have the biggest impact is shown through the Structure dimension. Together, these three dimensions form the conceptual framework of the knowledge base; besides defining its boundaries, they also indicate categories where effort can be directed in future research.

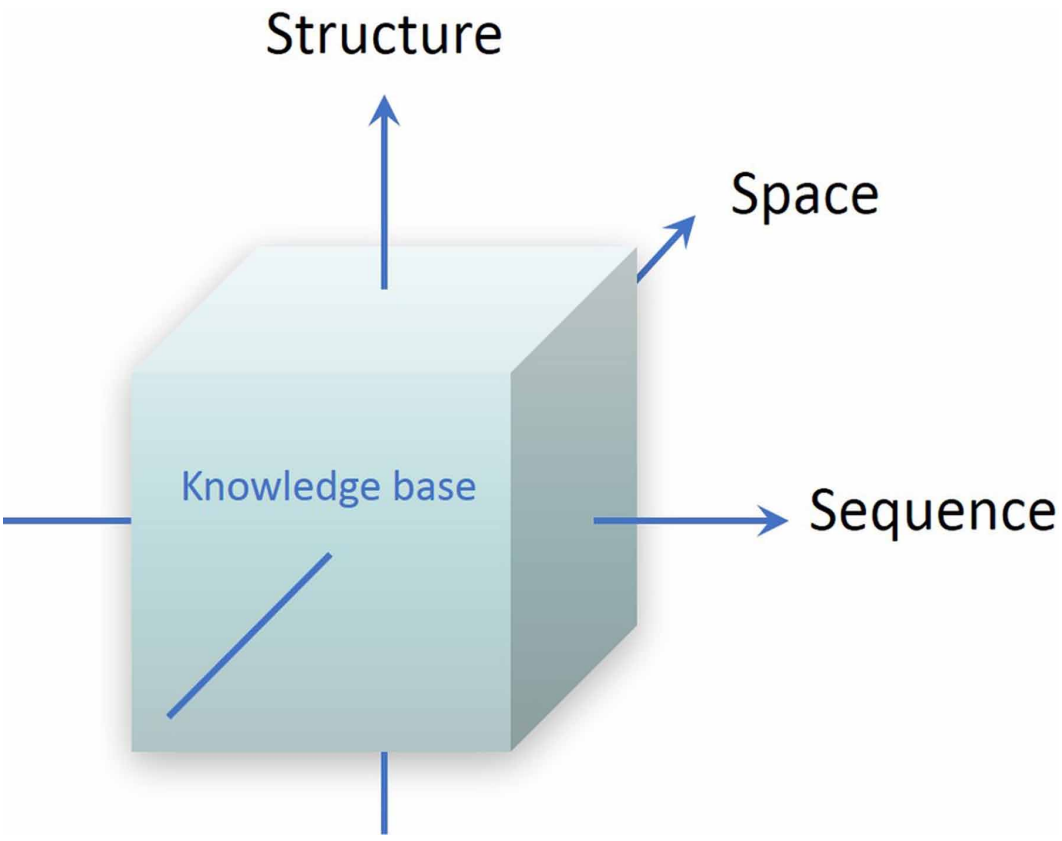
3. RESEARCH TOOLS AND METHODS

3.1 Origin of Search Standards and Definitions

To surface and analyze the status of applying collaborative learning in e-learning, this study takes Scopus as its search base. As the largest citation database of academic literature in the world, the Scopus database encompasses a very broad range of subjects, so can be considered a representative

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Figure 1. 3S Conceptual Framework of e-CL Knowledge Base



global index. Using the search terms Collaborat* AND e-Learning OR Online learning as keywords, with no beginning date restrictions through and including 2019, a search of the literature turns up 8,610 entries. The literature types originally included comprise: Article, Book Chapter, Book, Conference Paper, Review, Editorial, Article in Press, and so on. Each type of literature has varying degrees of contribution, but in order to elevate the degree of rigor for analysis of the literature, it was decided to narrow the sample to include only complete publications. This results in 8,575 pieces of data comprising six types of literature being included in the sample for analysis: Article (2,186), Book Chapter (355), Book (37), Conference Paper (5,579), Conference Review (296), and Review (122).

As the data was being organized, it was discovered that when some countries and parts of names were entered into the Scopus database, there were some discrepancies in spelling or expression that resulted in differences. In order to improve accuracy in the succeeding analysis, these discrepancies were resolved by unifying the expressions with the same meaning. For example, among country names, USA and United States were combined; in journal names, Computer and Education and Computer & Education were combined; within keywords, CSCL, Computer Supported Collaborative Learning and Computer-Supported Collaborative Learning were combined, and so forth.

3.2 Data Analysis

The bibliometric method is a supplemental tool for one type of traditional literature review. This analysis can provide researchers with useful information about a field of research, such as: What are the important publications and who are the important authors in this field? What is the geographic structure of this field? What are the trends and directions of research in each country? It can also

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provide additional information on the structured literature review. Compared to a summary of the literature, scientific landscape mapping provides a macro perspective. Traditional literature review provides depth, while the bibliometric method can process a range of research literature that numbers in the hundreds, or even thousands.

VOS scientific landscape mapping is one type of tool used in quantitative method design, and uses several data analysis methods, such as simple descriptive statistics to describe the basic time and space characteristics of the literature (such as size, geographic distribution, theme distribution) and knowledge base dynamics (such as time series analysis) (Hallinger, 2018a, 2018b; Hallinger & Bryant, 2013).

In bibliometric research, distance-based maps are maps in which the distance between two items reflects the strength of the relationship between the items. A smaller distance generally indicates a stronger relationship and will gather the items into a cluster. Clusters do not need to exhaustively cover all items in a map; there may be items that do not belong to any cluster. A cluster is a set of items included in a map; items may be grouped into clusters. (Van Eck & Waltman, 2010, 2013, 2017).

4. RESULTS

The results of analyzing the e-CL knowledge base are presented in this section according to the three research questions of this study.

4.1 Volume of Literature and Growth Trajectory

Employing the search criteria described above, 8,575 related pieces of literature are retrieved from the Scopus citation index, from the earliest by Hiltz (1988) in which computer software was used to assist in teaching and collaborative learning in setting up a Virtual Classroom; when this was compared to traditional classroom, totally online and mixed mode, students generally felt that the virtual classroom provided a fine learning experience. The succeeding decade saw only scattered publications in this field; only in 2000 did the trend begin to accelerate. In the three years from 2017 to 2019, a period of explosive growth in this field of research, there were a total of 2,165 papers, accounting for 25% of the total, with the most in 2019, at 779 (Figure 2).

The most frequently appearing author keywords are retrieved from an author keyword co-occurrence analysis applied to the 8,575 papers, the results of which can produce an author keyword map (Figure 3). Filtering out the keywords used in this study's search (collaborat*, e-learning, online learning), common related terms in this field (education, learning), and closely related phrases (distance learning, CSCL, computer-supported collaborative learning), the most commonly used keywords in order of frequency are: mobile learning, blended learning, higher education, Web2.0, collaborative filtering, virtual reality, knowledge management, MOOC, wiki, and learning analytics. The author keyword map uses colors to illustrate the popularity of these keywords over recent years (Figure 3). The commonly appearing keywords Web2.0, knowledge management and wiki seldom appeared prior to 2010, only suddenly soaring in popularity after 2010. Then there was a gradual shift to mobile learning, blended learning, higher education and collaborative filtering. Around 2015, virtual reality, MOOC and learning analytics rose in popularity and it can be seen that these themes appear frequently in the recent years of e-CL literature.

4.2 Geographic Distribution

The contributions from countries all over the world to e-CL research is not evenly distributed; papers from the United States and the United Kingdom account for 25% of the total for countries with over 100 publications each (Figure 4). These 26 countries can be considered the major contributors to this research, accounting for 90% of the total.

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Figure 2. Growth Trajectory of e-CL Knowledge Base

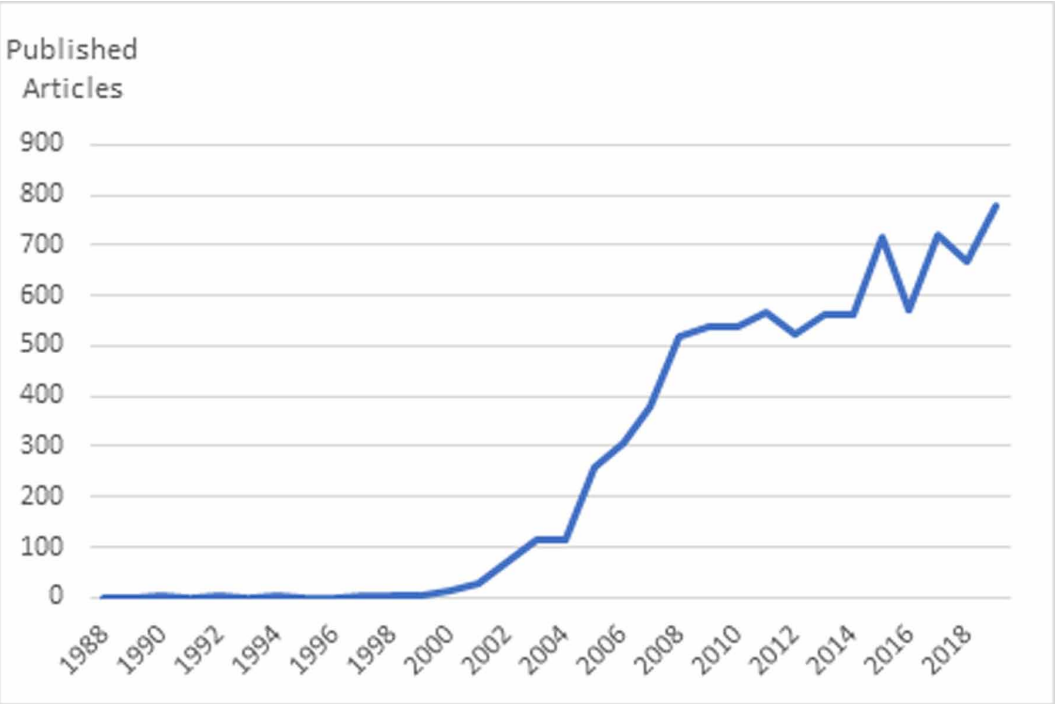
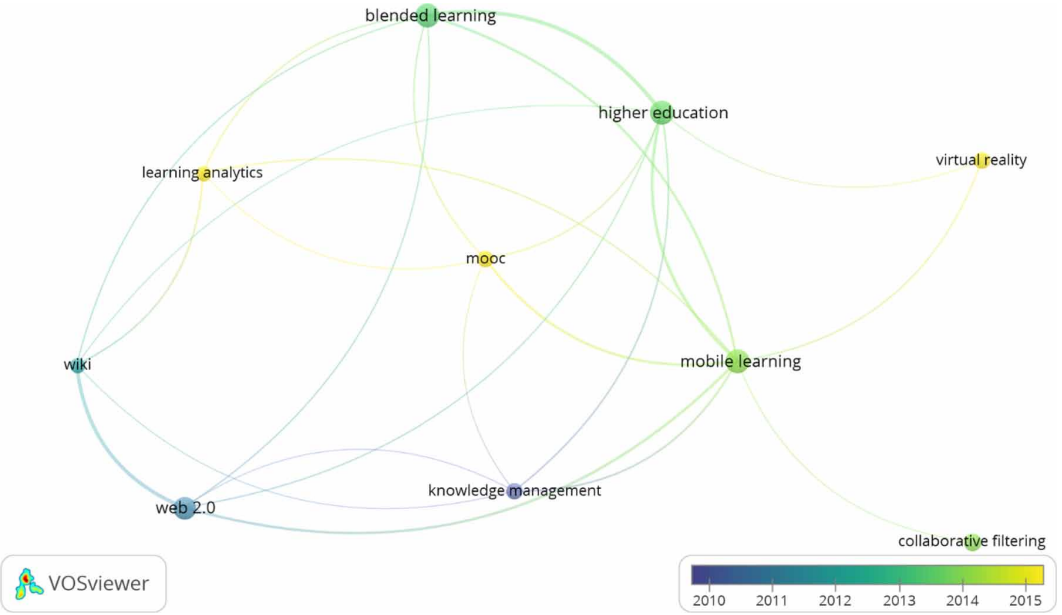


Figure 3. Author Keyword Map of e-CL Knowledge Base



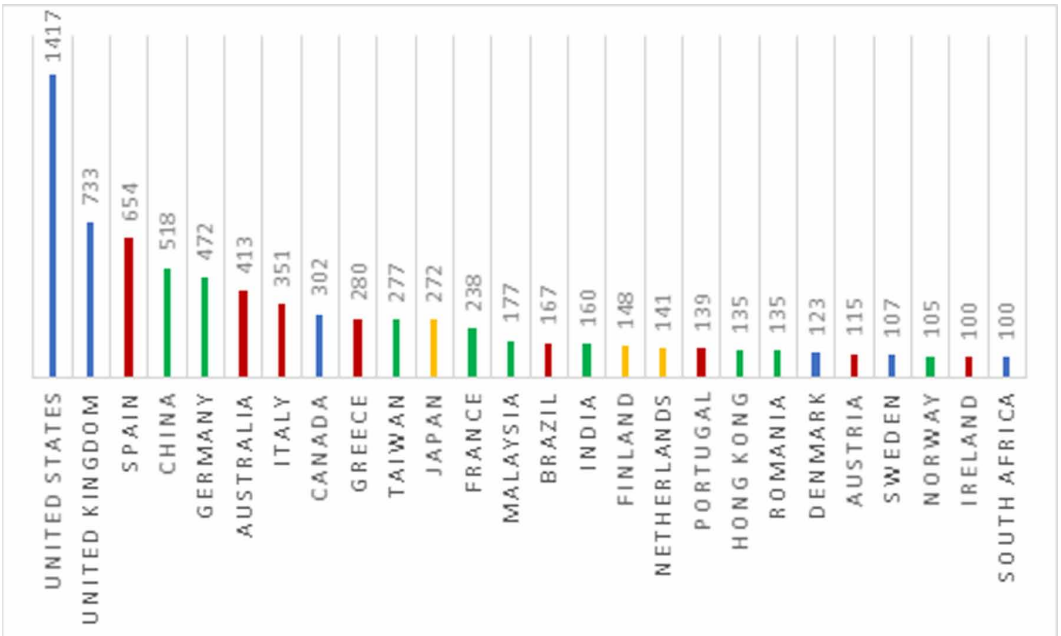
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A citation analysis on the literature from these 26 nations, taking into account the frequency and relationship of literature mutually cited, divides them into four clusters (Figure 5), which are shown below:

Cluster 1: China, France, Germany, Hong Kong, India, Malaysia, Norway, Romania, Taiwan
Cluster 2: Australia, Austria, Brazil, Greece, Ireland, Italy, Portugal, Spain
Cluster 3: Canada, Denmark, South Africa, Sweden, United Kingdom, United States
Cluster 4: Japan, Finland, Netherland

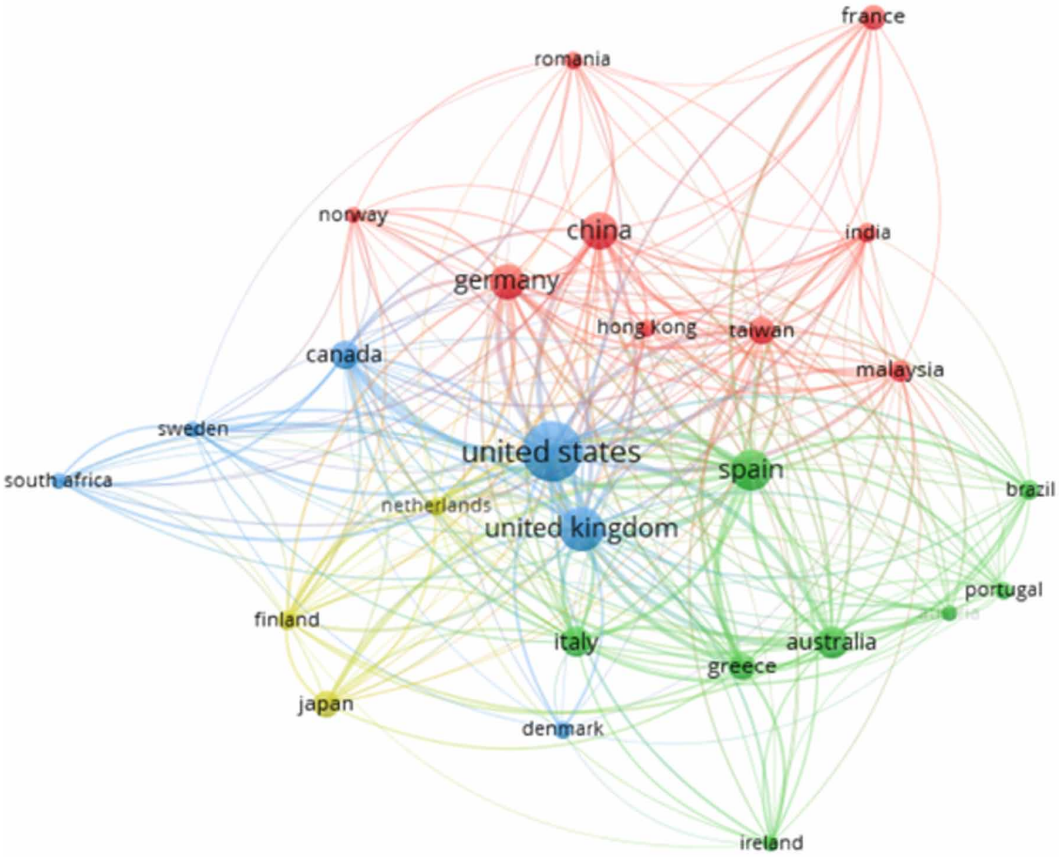
It can be clearly seen that these clusters are not formed from geographic proximity; for example, the Scandinavian nations of Norway, Denmark and Finland are found in separate clusters, while geographically distant Finland, Japan and the Netherlands are in the same cluster. Following, within the clusters, co-occurrence analysis was applied to author keyword to find the top 20 keywords in each cluster. After the filtering out the keywords used in this study's search, common related terms in this field and closely related phrases, the author keywords common to all of the clusters are: mobile learning, blended learning, and higher education, which can be seen as the focus of e-CL-related research being carried out all over the world. Moreover, virtual reality and web2.0 are common author keywords to clusters 1, 2 and 3, while learning analytics is common to clusters 2, 3 and 4 and knowledge management is a common author keyword to clusters 1 and 2, illustrating the common topics of research among these clusters. The most frequently appearing author keywords in individual clusters are collaborative filtering, MOOC, learning management system, wiki, and cloud computing in cluster 1; learning design, lifelong learning and assessment in cluster 2; social media and technology in cluster 3; and gamification, Moodle and project-based learning in cluster 4.

Figure 4. Global Distribution of e-CL Literature



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Figure 5. Country Cluster Map of e-CL Knowledge Base



4.3 Influential Journals, Literature and Authors

4.3.1 Journals with Impact

Among all the papers and citations searched in this research, the citation count amounts to a grand total of 83,970; from among these diverse sources, the 20 most frequently occurring publication sources account for at least 355 citations. Computers in Human Behavior heads up a cluster of nine journals, wherein 92 of the e-CL papers are published, cited a total of 5,040 times. Computers & Education heads up a cluster of seven journals, wherein 271 of the e-CL papers are published, cited a total of 7,021 times. Journal of Computer Assisted Learning heads up a cluster of four journals, wherein 54 of the e-CL papers are published, cited a total of 2,489 times. According to published papers and citation frequency, the average citations for papers in the Computers & Education cluster exceeds that in the other two groups by at least 400 citations. The journal in this cluster with the highest number of publications is British Journal of Educational Technology, with 69 papers, far more than those of any journal in the other clusters. From this, it can be known that, in the area of e-CL research, journals in the second cluster have greater impact (Table 1).

Citation data shows that among the 20 journals (Table 2) with the highest correlation of citation frequency, nine of these are not included in the list of those publishing the 8,575 papers in this study, which illustrates that e-CL theory originates from specializations in other fields. Thus, in pushing this exploration deeper through the journal keywords in the three clusters of journals, the researchers found distribution in other related fields.

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Table 1. Journal Cluster of the e-CL Knowledge Base

Computers in Human Behavior	Computers & Education	Journal of Computer Assisted Learning
International Journal of Computer-Supported Collaborative Learning	British Journal of Educational Technology	Educational Technology & Society
Educational Technology Research and Development	The Internet and Higher Education	Communications of the ACM
Educational Psychologist	Journal of Asynchronous Learning Networks	MIS Quarterly
Learning and Instruction	Australasian Journal of Educational Technology	
Review of Educational Research	Distance Education	
Educational Researcher		
Journal of the Learning Sciences		
Instructional Science		

Table 2. The 20 Most Important Publications

Publications	Cluster	citations	total link strength
Computers in Human Behavior	1	1384	11769
International Journal of Computer-Supported Collaborative Learning	1	549	4358
Educational Technology Research and Development	1	528	3830
Educational Psychologist	1	466	3719
Learning and Instruction	1	462	4074
Review of Educational Research	1	456	3346
Educational Researcher	1	421	2086
Journal of The Learning Sciences	1	419	2446
Instructional Science	1	355	2944
Computers & Education	2	3571	22563
British Journal of Educational Technology	2	1148	8027
The Internet and Higher Education	2	704	5733
Journal of Asynchronous Learning Networks	2	427	3104
Australasian Journal of Educational Technology	2	407	2946
Distance Education	2	402	2560
Internet and Higher Education	2	362	3229
Journal of Computer Assisted Learning	3	813	6112
Educational Technology & Society	3	706	4554
Communications of The Acm	3	544	1441
Mis Quarterly	3	426	2295

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In the nine-journal cluster headed by Computers in Human Behavior, wherein 92 of the e-CL papers are published, there are a total of 354 author keywords. The 10 most highly correlated appear at least four times in papers in this cluster. In the seven-journal cluster headed by Computers & Education, wherein 271 of the e-CL papers are published, the author keywords total 419. The 10 most highly correlated appear at least 13 times in papers in this cluster. In the four-journal cluster headed by Journal of Computer Assisted Learning, wherein 54 of the e-CL papers are published, the total author keywords count is 203. The 10 most highly correlated appear at least two times in papers in this cluster. Among the three clusters, aside from the keywords of collaborative learning and e-learning, higher education and mobile learning are commonly occurring keywords in cluster 1 and cluster 3 (Table 3), illustrating that higher education and mobile learning are important themes in the e-CL field of research.

Table 3. Keywords of the Three Clusters in e-CL Knowledge Base

Computers in Human Behavior	Computers & Education	Journal of Computer Assisted Learning
social networks	interactive learning environments	collaborative filtering
***higher education**	computer-mediated communication	recommender systems
***mobile learning**	teaching/learning strategies	*higher education
learning analytics	learning communities	*mobile learning
social media	pedagogical issues	ubiquitous learning

Note: *indicates keywords that are duplicated in different clusters

4.3.2 Literature with Impact

The 8,575 publications in this study are also analyzed for the 20 most frequently cited, as the most important papers in the e-CL field; each is cited at least 197 times, as shown in the following table. Among them, 10 are published in the Computers & Education journal, including the five most frequently cited papers, which illustrates the importance of this journal in the e-CL field (Table 4).

The 8,575 publications from the Scopus database studied in this research have a total of 27 research field correlations. Those accounting for over 1% of the related publications belong to the nine research fields of Computer Science; Social Sciences; Engineering; Mathematics, Business, Management and Accounting; Decision Sciences; Medicine; Arts and Humanities; and Psychology (Table 5). Among them, most are in Computer Science, followed by Social Sciences. Computer Science, Engineering, Mathematics, which, along with Arts and Humanities, are part of the STEAM disciplines, illustrating that e-CL research in STEAM fields is very popular; however, there is relatively little research in Technology.

4.3.3 Authors with Impact

Among the 8,575 papers analyzed in this research, there are 17,903 authors. The top 20 cited authors each published at least 15 papers related to e-CL. Analysis of the correlation between authors' mutual citations and co-authoring results in forming the author clusters of: [Xhafa, Caballé, and Barolli], [Li, Li and Chen], [Daradoumis, Tsiatsos and Dimitriadis] and [Temperini and Sterbini]. Being clustered together shows that there is significant correlation between them regarding co-authoring or mutual citations, indicating that their areas of research are closely related. Although nine authors – Gillet, Kim, Okamoto, Rummel, Trausan-Matu, Wang J., Wang X., Zhang J. and Zhang X. – collectively published more e-CL research papers than others, little correlation is found between them in regard to citations or collaborative publication.

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Table 4. Top 20 Important Publications in e-CL Knowledge Base

Publication	Citations	Title	Contributions / Focus in e-CL	Source
Cheung R. (2013)	254	Predicting user acceptance of collaborative technologies: An extension of the technology acceptance model for e-learning	The ability to share information in the collaborative learning environment is found to influence intention and behavior toward the Google Applications platform.	Computers & Education
Mikropoulos T.A. (2011)	286	Educational virtual environments: A ten-year review of empirical research (1999–2009)	This study is a ten-year critical review of empirical research on the educational applications of Virtual Reality. Findings show that collaboration and social negotiation are not only limited to the participants of an EVE, but exist between participants and avatars, offering a new dimension to computer assisted learning.	Computers & Education
Jones C. (2010)	387	Net generation or Digital Natives: Is there a distinct new generation entering university?	This study found that new technology was often used in ways that did not fully correspond with the expectations that arise from the Net generation and Digital Natives theses. The authors find that the generation is not homogenous in its use and appreciation of new technologies and that there are significant variations amongst students that lie within the Net generation age band.	Computers & Education
Macfadyen L.P. (2010)	437	Mining LMS data to develop an "early warning system" for educators: A proof of concept	This paper confirms and extends how higher education institutions could harness the predictive power of Learning Management System (LMS) data to develop reporting tools that identify at-risk students and allow for more timely pedagogical interventions. A growing body of research on teaching and learning in higher education emphasizes the adoption of learner-centered pedagogical approaches and the critical importance of encouraging student–student interaction/collaboration, promoting active learning, providing prompt and detailed feedback, and stressing the need for time on task, whilst also respecting the diverse modes of learning.	Computers & Education
Paechter M. (2010)	271	Students' expectations of, and experiences in e-learning: Their relation to learning achievements and course satisfaction	This study investigates how different facets of students' expectations and experiences are related to perceived learning achievements and course satisfaction and relevance. Students' achievement goals were the best predictors for success and ranked higher than other course characteristics. Furthermore, self-regulated and collaborative learning were related to learning achievements.	Computers & Education
Frohberg D. (2009)	223	Mobile Learning projects – a critical analysis of the state of the art	This paper provides a critical analysis of Mobile Learning projects published before the end of 2007. Discussion of the MOOP (Mattilaand Fordell 2005) project explains how the system is explicitly designed as a tool to support cooperative learning in school classes. The purpose of the system is to bring different observations from nearby surroundings into the classroom. The learners can create photos, videos and voice annotations with a GPS location tag, exchange them and work with them, collaboratively generating their own knowledge.	Journal of Computer Assisted Learning

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Table 4. Continued

Publication	Citations	Title	Contributions / Focus in e-CL	Source
Hoic-Bozic N. (2009)	198	A Blended Learning Approach to Course Design and Implementation	This paper describes the use of the blended e-learning model, which is based on a mixture of collaborative learning, problem-based learning and independent learning. Supporting collaborative and project-oriented activities promotes students' motivation for learning and establishes learning as an active and interactive process. A survey conducted in the end of the course showed that students were satisfied with the pedagogical approach, and their academic achievements were also better than expected. Particularly important is that the dropout rate was greatly diminished, which could be related to students' satisfaction with the support they received from the instructor and the system.	IEEE Transactions on Education
De Lucia A. (2009)	242	Development and evaluation of a virtual campus on Second Life: The case of Second DMI	The study conducted an experiment involving university students aiming at evaluating Second Life synchronous distance lectures in the virtual campus learning environment. The evaluation has been conducted considering that, in a 3D multi-user virtual environment, learning is strongly related to the user perception of belonging to a learning community, as well as to the perception of awareness, presence and communication. The results of the evaluation are very positive.	Computers & Education
So H.-J. (2008)	438	Student perceptions of collaborative learning, social presence and satisfaction in a blended learning environment: Relationships and critical factors	The purpose of this study was to examine the relationships of the students' perceived levels of collaborative learning, social presence and overall satisfaction in a blended learning environment. The analysis of quantitative data indicated that student perceptions of collaborative learning have statistically positive relationships with perceptions of social presence and satisfaction. This means that students who perceived high levels of collaborative learning tended to be more satisfied with their distance course than those who perceived low levels of collaborative learning. Similarly, students with high perceptions of collaborative learning perceived high levels of social presence as well.	Computers & Education
Monahan T. (2008)	197	Virtual reality for collaborative e-learning	This article presents research in this area and the resulting development of CLEV-R, a Collaborative Learning Environment with Virtual Reality. The mobile version of CLEV-R provides access to course announcements, communication with other users and the ability to download course notes through an interactive 3D environment.	Computers & Education
Motiwalla L.F. (2007)	495	Mobile learning: A framework and evaluation	This project explores the extension of e-learning into wireless/handheld computing devices with the help of a mobile learning framework. A prototype application was developed to link wireless/handheld devices to three course websites. The results from this exploratory study provide a better understanding on the role of mobile technology in higher education.	Computers & Education

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Table 4. Continued

Publication	Citations	Title	Contributions / Focus in e-CL	Source
De Laat M. (2007)	218	Investigating patterns of interaction in networked learning and computer-supported collaborative learning: A role for Social Network Analysis	The focus of this study is to explore the advances that Social Network Analysis can bring, in combination with other methods, when studying Networked Learning/Computer-Supported Collaborative Learning (NL/CSCL). The study presents a general overview of how SNA is applied in NL/CSCL research and illustrates how this research method can be integrated with existing studies on NL/CSCL as a way to synthesize and extend our understanding of teaching and learning processes in this context.	International Journal of Computer-Supported Collaborative Learning
Kamel Boulos M.N. (2006)	667	Wikis, blogs and podcasts: a new generation of Web-based tools for virtual collaborative clinical practice and education	This study discusses how effectively deployed wikis, blogs and podcasts could offer a way to enhance students', clinicians' and patients' learning experiences, and deepen levels of learners' engagement and collaboration within digital learning environments. Of particular importance is research into novel integrative applications, to serve as the "glue" to bind the different forms of Web-based collaboration tools synergistically in order to provide a coherent learning experience.	BMC Medical Education
Hernańdez-Leo D. (2006)	214	COLLAGE: A collaborative Learning Design editor based on patterns	This paper introduces Collage, a high-level IMS-LD compliant authoring tool that is specialized for CSCL (Computer-Supported Collaborative Learning). Nowadays CSCL is a key trend in e-learning since it highlights the importance of social interactions as an essential element of learning. An example of a Learning Design (LD) that can be created using Collage is illustrated in the paper. Preliminary evaluation results show that teachers with experience in CL but without LD knowledge, can successfully design real collaborative learning experiences using Collage.	Journal of Educational Technology & Society
Ruiz J.G. (2006)	880	The Impact of E-Learning in Medical Education	The authors provide an introduction to e-learning and its role in medical education by outlining key terms, the components of e-learning, the evidence for its effectiveness, faculty development needs for implementation, evaluation strategies for e-learning and its technology, and how e-learning might be considered evidence of academic scholarship. Innovations in e-learning technologies point toward a revolution in education, allowing learning to be individualized (adaptive learning), enhancing learners' interactions with others (collaborative learning), and transforming the role of the teacher.	Academic Medicine
Paavola S. (2005)	278	The Knowledge Creation Metaphor – An Emergent Epistemological Approach to Learning	This paper reviews three approaches to knowledge-creation, i.e., Bereiter's knowledge-building, Engestr□m's expansive learning, and Nonaka and Takeuchi's organizational knowledge-creation. It gives a concise analysis of the trialogical character of the knowledge-creation approach and illustrate how the third metaphor may be applied at the school level.	Science & Education

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Table 4. Continued

Publication	Citations	Title	Contributions / Focus in e-CL	Source
Sadiq S.W. (2005)	209	Specification and validation of process constraints for flexible workflows	This paper presents a foundation set of constraints for flexible workflow specification. These constraints are intended to provide an appropriate balance between flexibility and control. The constraint specification framework is based on the concept of "pockets of flexibility" which allows ad hoc changes and/or building of workflows for highly flexible processes. The workflow concepts have been used to support non-coordinative (collaborative) processes. speaks of ad hoc workflows, where the process cannot be completely defined prior to execution. It also discusses the coordination of collaboration intensive processes.	Information Systems
Chen C.-M. (2005)	307	Personalized e-learning system using Item Response Theory	To determine an appropriate level of difficulty parameter for the course materials, this study proposes a collaborative voting approach for adjusting course material difficulty. Experiment results show that applying Item Response Theory to Web-based learning can achieve personalized learning and help learners to learn more effectively and efficiently. In the proposed system, the difficulty of the course material can be dynamically tuned using the proposed collaborative voting approach after learners offer feedback.	Computers & Education
Sharples M. (2002)	215	The Design and Implementation of a Mobile Learning Resource	This article describes the theory-informed design, implementation and evaluation of a handheld learning device. It is intended to support children to capture everyday events such as images, notes and sounds, to relate them to web-based learning resources, and to organize these into a visual knowledge map. Learning is a continual conversation: with the external world and its artefacts, with oneself, and also with other learners and teachers. The most successful learning comes when the learner is in control of the activity, able to test ideas by performing experiments, asking questions, collaborating with other people, seeking out new knowledge, and planning new actions.	Personal and Ubiquitous Computing
David Merrill M. (2002)	691	First principles of instruction	This paper identifies prescriptive principles that are common to the various theories of instruction. Representative instructional design theories include: Star Legacy by the Vanderbilt Learning Technology Center, 4-Mat by McCarthy, instructional episodes by Andre, multiple approaches to understanding by Gardner, collaborative problem solving by Nelson, constructivist learning environments by Jonassen, and learning by doing by Schank. It is concluded that, although they use a wide variety of terms, these theories and models do include fundamentally similar principles.	Educational Technology Research and Development

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Table 5. Important Research Fields in e-CL

Research Field	No. of Papers	Percentage of Total Papers
Computer Science	**6,347**	42.08%
Social Sciences	**4,106**	27.22%
Engineering	1,747	11.59%
Mathematics	945	6.27%
Business, Management and Accounting	374	2.48%
Decision Sciences	363	2.41%
Medicine	272	1.81%
Arts and Humanities	173	1.15%
Psychology	150	1%

From among those mentioned above who have published at least 15 papers in this field and have the highest number of citations in this field, the top 20 authors are divided into four clusters based on the similarity of their publications. Using Author as the search filter, author keyword co-occurrence analysis is carried out once again using VOS software on all publications in this field for each cluster of authors. Once the keywords of collaborative learning and e-learning set for this research are removed, the result of the analysis is a list of the five most commonly occurring keywords for each cluster of authors, from which the research topics receiving most attention by each author cluster can be seen (Table 6).

5. DISCUSSION AND CONCLUSION

5.1 Research Findings

The major findings from the bibliometric analysis of the 8,575 publications on the e-CL theme in this research are: (1) the focus on e-CL-related research continues to grow; (2) research related to e-CL is not evenly distributed geographically in terms of volume of publications and the geographic distribution is not clustered according to proximity; (3) the journals with the most impact in e-CL research are: Computers in Human Behavior, Computers & Education and Journal of Computer Assisted Learning; (4) the most impactful authors in e-CL research are: Barolli, Caballé, Chen,

Table 6. Keywords of Four Author Clusters in the e-CL Knowledge Base

Xhafa F., Caballé S., Barolli L.	Li X., Li Y., Chen W.	Daradoumis T., Tsiatsos T., Dimitriadis Y.	Temperini M., Sterbini A.
e-assessment	blended learning	affective feedback	adaptive e-learning
emotion awareness	collaboration	emotion	assessment
information security	collaborative filtering	emotion awareness	reputation system
trustworthiness	collaborative virtual maintenance	orchestration	student modeling
virtualized collaborative sessions	semantic wiki		zone of proximal development

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Daradoumis, Dimitriadis, Li, Li, Sterbini, Temperini, Tsiatsos and Xhafa; (5) 10 of the 20 papers with the highest impact in e-CL research are published in Computers & Education, including those with the top five number of citations.

5.2 Research Limitations

The citation values for the papers found in the search of this study vary slightly, which may be due to a linear impact on a small portion of the data or that when the paper was entered into the Scopus database, explanation of some data was not provided. For example, regarding the country of origin for publications, 351 of them are marked as Undefined, which results in slight discrepancies in data calculation. As these slight discrepancies do not affect the analysis of trends in this research, no deeper analysis of them is addressed in this study.

5.3 Conclusion

5.3.1 e-CL Research is Currently Primarily Targeted on Tertiary Education

Through author keyword analysis it is found that Higher Education appears very often in this whole knowledge base and is a high frequency keyword in each of the clusters of nations, which illustrates that e-CL research is developing vigorously in higher education, while much less in K-12. In terms of journals with impact in e-CL research, publications related to higher education are mostly published in the nine-journal cluster headed by Computers in Human Behavior and the four-journal cluster headed by Journal of Computer Assisted Learning.

5.3.2 e-CL Research is Extensively Applied in Specific Subjects, not to Minor Subjects

Through the source journal analysis of the 8,575 publications, it can be seen that journals in the fields of Sciences, Engineering, Mathematics, Business, Management and Accounting all contain e-CL research, illustrating that the research topic of e-CL is being extensively discussed in all fields. Combined, source journals in the Computer Science and Social Sciences fields account for 70% of the total. However, as some e-CL topics are essentially a combination of topics from these two fields, it cannot simply be inferred that e-CL is investigated most in Computer Science or Social Sciences subject applications.

5.3.3 e-CL Research Actively Integrates New Technology Applications

In the analysis of Author keyword in this Knowledge base, it is found that mobile learning, web2.0, virtual learning, MOOC and wiki appeared many times, indicating that these technologies are commonly combined in today's e-CL research. Moreover, this type of research is concentrated in the cluster of China, France, Germany, Hong Kong, India, Malaysia, Norway, Romania, and Taiwan. Among them, virtual learning and MOOC are research topics that have emerged in recent years.

5.3.4 Research Topic Borders

Citation analysis shows that there are cluster relationships among nations which are not simply formed based on geographic proximity. Through author keyword co-occurrence analysis, using the 'country' factor, it can be seen that national borders are blurred in this research topic, while the 'journal' factor exhibits a much stronger distinction. The keywords of the most prolific authors show little overlap between author clusters. These findings illustrate that regions of neighboring countries do not necessarily have similar research topic trends and research topics are frequently cross-border, but the topical focus of researchers and journals is much more long-term and stable.

Employing the 3S conceptual framework of Sequence, Space and Structure to analyze the e-CL knowledge base, it can be seen from the Sequence dimension that research in the e-CL field continues to draw much attention. Hot keywords after 2015 are virtual reality, MOOC, and learning analytics; however, the Sequence dimension also reveals that keywords continue to be hot for approximately 3-5

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years. So, the 2018-2019 literature also commonly includes the following keywords: gamification, active learning, engagement, deep learning, interactive learning environments, flipped classroom, and augmented reality. Although the current usage frequency of these themes in the knowledge bank in not yet considered significant, the changes in Sequence indicate that these themes have much potential for future development in the e-CL field. The speed of Internet development has virtually broken through the limitations of geographic space; analysis of the Space dimension reveals the research trends that each nation is partial to, providing researchers with literature sources for consideration in regard to topics of interest. Moreover, analysis of the Structure dimension within the e-CL field reveals topics of interest in high impact journals, research contributions of important literature and most frequently cited authors in the e-CL field, which provides explicit research references for e-CL researchers regarding topics attracting attention in these disciplines.

Conflicts of Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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Influencing Factors of Intention to Use Mobile Devices for Reading:

Moderation Effects of Perceived Sociality and Perceived Risk

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ABSTRACT

This research studied the continuous use intention of mobile reading users. The current study used the stimulus-organism-response model as a basis to build a causal model of mobile readers' continuous intention to use it. A questionnaire survey was conducted involving 327 users to obtain the current research data. Structural equation modeling was used to empirically test the relationship amongst variables in the conceptual model. Results are as follows. Perceived pleasure, perceived service, immersion, and perceived sociality had a significant positive effect on intention for continuous use. Particularly, perceived pleasure had a significant positive effect on immersion and perceived service. Perceived usefulness and perceptual interest had significant positive effects on perceived pleasure. Perceived sociality had a regulating effect on the relationship between perceived pleasure and intention for continuous use. Lastly, perceived sociality had a regulating effect on the relationship between immersion and intention for continuous use.

KEYWORDS

Mobile Devices for Reading, Perceived Pleasure, Perceived Risk, Perceived Society, Use

1. INTRODUCTION

The rapid development of mobile Internet technology has resulted in substantial changes in the way people read. The use of mobile devices for reading has become mainstream as a new reading mode on mobile platforms (Hellermann, Thorne & Fodor, 2017). In the field of reading promotion, the use of mobile devices for reading is becoming an indispensable part of the public's daily life (Wang, Chiu, Ho & Lo, 2016). The emergence of mobile devices for reading has brought both immense opportunities and challenges related to the promotion of digital reading. A 2019 report indicated that the number of active users of mobile devices for reading in China reached 350 million RMB. In addition, the market size reached 20.49 billion RMB, with a year-on-year increase of 22.4% (iResearch, 2020).

Although the use of mobile devices for reading occupies most of the market share, users consistently shift. While maintaining the growth of the use of mobile devices for the reading community, preventing the loss of mobile devices for reading users has become an important issue for digital reading promoters (Zhao & Ren, 2017). Therefore, the focus on the rapid development

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of the use of mobile devices for reading necessitates an analysis of the factors that influence the use of mobile devices for reading users' continuous use behaviors. In addition, we must promote the sustainable development of mobile devices for reading.

The use of mobile devices for reading refers to accessing, receiving, or downloading electronic resources through a wireless or mobile communication network for reading, particularly on a handheld portable device (Zhang, 2011). In the current study, the use of mobile devices for reading refers to the use of mobile terminals to read novels, magazines, and newspapers, among others, through Web pages and bookstore clients.

The use of mobile devices for reading improves the utilization of fragmentation time, which refers to the break period in the user's work and life that is affected by the diversification of needs (Nie et al., 2020). On the mobile Internet, fragmented time refers to the scattered times when people wait for elevators, buses, and subways, as well as getting in cars (Christensen & Knezek, 2018). Mobile Internet users can make full use of fragmented time to browse news, play games, and/or read e-books through mobile phones and tablets (Sarrab, Elbasir, & Alnaeli, 2016). Although these fragmented times are inconspicuous, they can produce important effects (Matronchik, Klyachin, & Khangulyan, 2018).

Variations in reading media have led to substantial changes in reading behaviors. A reading medium refers to the carrying and transmission of text based on text symbols, providing reading options rather than watching and listening. These materials are capable of direct physical contact with readers (i.e., books, newspapers, and magazines). They are based on paper-based printing techniques or digital information technology like computers, smart phones, and e-readers (Delgado, Vargas, Ackerman & Salmerón, 2018; Ghassemi et al., 2019). The reading medium mentioned in this paper refers to portable mobile devices, such as mobile phones and tablets, with a prompt off/on button (Crompton, 2013).

The rapid development of the use of mobile devices for reading has resulted in the emergence of characteristics in users' reading behaviors (Vasileiou, Hartley, & Rowley, 2009). Research on the use of mobile devices for reading behaviors has become a popular topic in the field of reading research. Others research hotspots include user preferences regarding mobile devices for reading, mobile device applications for reading in education, and factors influencing mobile device adoption behaviors for reading.

Previous studies on reading behaviors related to the use of mobile devices for reading (Konok, Pogány & Miklósi, 2017; Lin, 2014; Merga & Mat Roni, 2017; Park & Lee, 2019) found fragmented and segmented behaviors (Halevi, Moed & Bar-ILan, 2015; Nardi, 2016). Other studies determined that college students' use of mobile devices for reading differs in terms of gender, grade, subject, and region (Ciampa, Thrasher, Marston & Revels, 2013). Researchers investigated college students' use of mobile devices for reading by analyzing differences between e-reading and paper books in classroom learning (Halevi et al., 2015; Rasmusson, 2016).

Research on adoption behaviors has been based on the technology acceptance model, emphasizing the use of specific technologies and services from an attitude perspective. Users' perception of a specific technology or service is believed to impact on attitudes toward the technology or service, thereby impacting user intention. The perceptual factors emphasized by the technology acceptance model are perceived usefulness, perceived accessibility, and perceived pleasure (Tsai, Wang & Lu, 2011).

Research on the use of mobile devices for reading has introduced customer choice theory and built a value-based adoption model. Studies have determined that the value of the reading content on mobile devices is also a major factor affecting adoption intention (Wang et al., 2016). Technologies in mobile devices for reading (i.e., compatibility, visibility, usage scenarios, and initial trust in services) also impact adoption (Chen & Lin, 2016). Most studies only consider the perspective of a single user. They do not explore interactions among users as they use the technology or service.

A few studies have explored the impact of social factors on adoption behaviors, particularly how image has proven to impact adoption behaviors of existing users. This limitation necessitates

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additional research on the influence of social factors on continuous use behaviors related to the use of mobile devices for reading. Therefore, this study analyzes the social perspective of continuous use behaviors related to mobile devices for reading.

The present research also investigates and verifies the status of the use of mobile devices for reading and social apps from the perspective of investments in time and frequency of use. Accordingly, the SOR (Stimuli, Organism, Response) model is introduced to supplement existing research on the social impact of the continuous use of mobile devices for reading. A model of factors affecting the continuous use of mobile devices for reading is constructed. The relationship between social factors and continuous use behaviors is explored both directly and indirectly.

This article is divided into seven parts. The first introduces the research background, significance, and content of mobile devices for reading and user intentions. The second reviews the literature regarding the use of mobile devices for reading. The third constructs a research model for influencing factors on the use of mobile devices for reading and continuous use intention. It also proposes the research hypothesis. The fourth systematically details the variables and measurement items in the research model, determines the questionnaire and research methods, and explains the data analysis methods. The fifth analyzes and draws conclusions from the collected valid questionnaire samples. The sixth assesses and summarizes the research conclusions. The seventh includes research limitations and future research prospects.

2. LITERATURE REVIEW

2.1 Use of Mobile Devices for Reading

The rapid development of the use of mobile devices for reading has attracted the attention of an increasing number of researchers. Scholars in different fields have published research studies on this topic. Nardi (2016) conducted research on the relationship between mobile readers and learning processes. The results show that e-textbooks have been popularized in Italy, the United States, and other countries. Therefore, teachers should focus on the influence of students' age, knowledge, reading skills, attitudes, and preferences on learning processes and results. Factors like satisfaction, perceived usefulness, and perceived entertainment impact continuous use intention (Yang, 2015). Naumann and Salmerón (2016) explored the impact of navigation and comprehension on the use of mobile devices for reading ability. Their research showed that comprehension significantly influences the use of mobile devices for reading ability. A strong understanding, however, does not produce a good use of mobile devices for reading skills. Rasmusson (2016) used the multilevel analysis method to study the overall use of mobile devices on the reading levels of Swedish and Norwegian students. The results showed that the overall use of mobile devices for reading levels is related to the cultural resources of the two countries. In addition, parental pressure impacted the use of mobile devices for reading levels of the Norwegian students. However, the effect of parental stress on Swedish students' use of mobile devices for reading levels was not significant. Overall, research has explored the trend of the use of mobile devices for reading. Few studies have been conducted on continuous use behaviors regarding the use of mobile devices for reading.

2.2 SOR Theoretical Model

The SOR model is a new theoretical model based on the (stimulus-response) theory extended by Mehrabian, a well-known environmental psychologist. This model is widely used in psychology, management, and economics. Stimuli (S) refers to an individual's environment and changes caused by that environment. Organism (O) includes the individual as an organic whole, internalizing the environmental stimulus into the process of individual change. Response (R) is the result of the interaction of the individual environmental stimuli and internalization processes. These are mainly manifested in changes in individual behavior patterns (Arora, 1982). The SOR model explains the

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relationship among the environment, internalization, and user behavior, providing a theoretical foundation for studying changes in user behaviors in a new environment.

Zheng, Li, Shen and Zheng (2019) used SOR and grounded theories to analyze the factors impacting user churn regarding the use of mobile devices and reading service platforms. The results showed that the main factors affecting user churn are low cost of user conversion platforms and influence of user social circles. Presently, the SOR model can substantially explain the usage and reading behaviors of users of information systems and those in mobile environments, respectively. Fang and Zhang (2019) used the SOR model to explore the influencing factors and paths of churn behaviors of users of mobile devices for reading. The study demonstrated that reading habits and system quality impact users' feelings of disappointment, thereby causing user churn. Zhao and Ren (2017) also used the SOR model to build a causal model of the continuous use intention of users of mobile devices for reading services. The results indicate that perceived usefulness and satisfaction were important factors for users regarding the continuous use of mobile devices for reading.

In the field of users and mobile devices for reading, the rise of new technologies and changes in reading environments have led to a change in users' psychological cognition and behavior patterns. In fact, behavior patterns were mainly affected by environmental changes. The core concept of knowledge behavior is consistent. Therefore, the SOR theoretical model indicated the feasibility of exploring the continuous use behavior of users of mobile devices for reading.

3. MODEL CONSTRUCTION AND RESEARCH ASSUMPTIONS

3.1 Research Model

This study used the SOR theory as a basis in building a research model on mobile devices for reading users' continuous use behaviors. Additionally, the study explains the impact mechanism of continuous use behaviors from the perspective of stimulus internalization response. The current research combined prior studies and developments in the use of mobile devices for reading to divide stimuli into external and internal stimuli. External stimuli include stimuli for clients who use mobile devices for reading. The selected indicators include timeliness and credibility of user perception systems. By contrast, internal stimuli focus on users' situations, including perceived usefulness, perceived accessibility, and perceived interest in the use of mobile devices for reading.

Internalization refers to the psychological changes that occur when users interact with internal and external stimuli. Before users have a continuous intention to use, they may feel good about the use of mobile devices for reading. Therefore, the internally selected observation variables include a sense of immersion and/or pleasure.

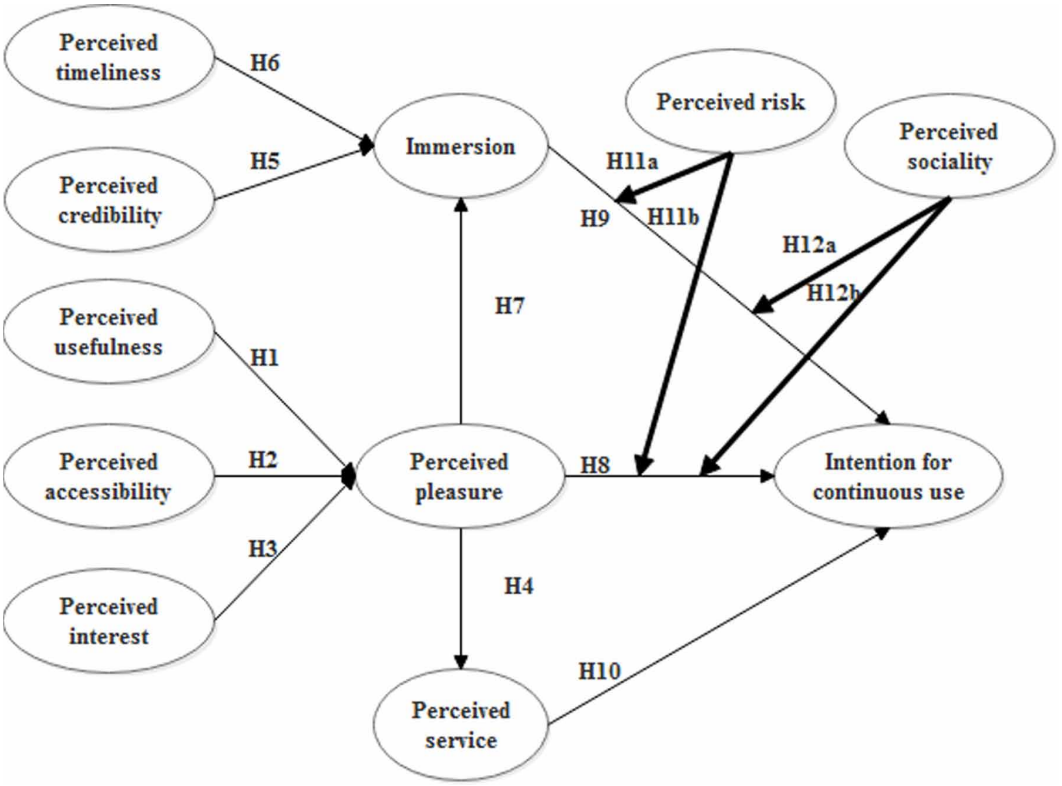
Response refers to affiliative and avoidance behaviors. This research studied the continuous use intention of users of mobile devices for reading, which is a specific manifestation of affiliative behavior. Therefore, the response considers affiliative behavior rather than avoidance. Figure 1 shows the specific research model, detailed in the following sections.

3.2 Research Hypothesis

Perceived pleasure is the user's level of happiness when interacting with the environment. This happiness is an unconscious psychological experience. When the environment or interaction scenario changes, the user's perceptions also change. Li and Chen (2019) studied the application of virtual reality technology in the tourism market. Their results showed that perceived accessibility and usefulness positively affect both perceived pleasure and travel intentions. Pe-Than, Goh and Lee (2014) believed that users' performance in computing games significantly affects perceived pleasure. Rodrigues, Oliveira and Costa (2016) investigated user experience in e-banking, finding a positive interaction between accessibility and perceived pleasure. Moghavvemi, Sharabati, Paramanathan and

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Figure 1. Research model



Rahin (2017) held that perceived usefulness significantly affects perceived pleasure when students share knowledge.

This study formulated the following hypotheses based on these research conclusions:

H1: Perceived usefulness significantly affects perceived pleasure.
H2: Perceived accessibility significantly affects perceived pleasure.
H3: Perceived interest significantly affects perceived pleasure.
H4: Perceived pleasure significantly affects perceived service.

The field of information systems uses the immersion theory to study users' virtual worlds (Cha, Koo, Kim & Hong, 2019; Tabrizian, Baran, Smith & Meentemeyer, 2018), instant messaging, online shopping (Lombart et al., 2019; Sinesio et al., 2019), and online learning (Cheng & Tsai, 2019; Meyer, Omdahl & Makransky, 2019). Perceived complexity, interactivity, and sociality can affect immersion. Immersion has also been used to study the behavior of mobile users. Jung, Begona and Sonja (2009) believed that the quality of mobile television content significantly impacts users' immersion. Tao & Yaobin (2011) showed that network externalities affect the immersive experience of instant messaging users. The use of mobile devices for reading is a category in virtual worlds. Users are highly concentrated in the use of mobile devices for reading, falling into the immersion experience process. In the era of the mobile Internet, an increasing number of service providers for mobile devices for reading have leveraged the immersion concept to provide extensive experiences for their users.

This study formulated the following hypotheses based on these research conclusions:

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H5: Perceived credibility has a significant positive effect on immersion.
H6: Perceived timeliness has a significant positive effect on immersion.
H7: Perceived pleasure has a significant positive effect on immersion.

Perceived pleasure is a dimension of perceived experience. Users' perceived pleasure affects the experience, and users' experience directly affects continuous use intention. Many studies have analyzed perceived pleasure as a factor affecting the willingness to continuously use information systems. Childers, Carr, Peck and Carson (2001) believed that perceived pleasure plays a key role in explaining and predicting Internet user behavior. Both Thong, Hong and Tam (2006) and Zhai, Sun abd Li (2019) used the perspective of user reviews to study the continuous use intention of mobile users on the Internet. The research considered perceived pleasure among the factors affecting continuous use intention, confirming that perceived pleasure significantly affects user satisfaction and continuous use intention. Perceived pleasure has also been seen to significantly affect the continuous intention to use mobile data services (Alalwan, Baabdullah, Rana, Tamilmani & Dwivedi, 2018; Kim, 2010; Sarosa, 2019).

Immersion refers to a state in which people are completely engaged in an activity and disregard the surrounding environment (Csikszentmihalyi, 1975). Initially, immersion was applied in the field of psychology. Recently, it has been introduced into the Internet and information systems. Kim and Thapa (2018) found that immersion has a significant impact on consumers' intention to purchase and repurchase goods, while Leung (2020) explained that immersion and purchase intention have a positive correlation on consumers as they browse Websites.

This study formulated the following hypotheses based on these research conclusions:

H8: Perceived pleasure has a significant positive effect on intention for continuous use.
H9: Immersion has a significant positive effect on intention for continuous use.
H10: Service quality has a significant positive effect on intention for continuous use.

Continuous use behavior is the follow-up development of adoption behavior, thereby necessitating the consideration of influencing factors in adoption behavior. In the study of adoption behavior, social factors play a significant role in continuous use behavior. Compared with adoption behavior, continuous use can better reflect interaction among users. The existing research has confirmed that social factors influence the continued use of mobile instant messaging (Ogara, Koh & Prybutok, 2014; Sheer & Rice, 2017). Klobas, McGill and Wang (2019) believed that perceived risk has a considerable impact on users' continuous intention to use smart homes. The risks perceived by users when using search engines will have a positive impact on behavioral intentions (Chao, Chang, Wu, Lin, & Chen, 2016). Fang, Tang, Yang and Peng (2019) found that social interaction has a positive effect on the intention of learning lessons. Few studies have been conducted on the continuous use behavior of the use of mobile devices for reading; therefore, the impact of social factors is minimal.

This study formulated the following hypotheses based on these research conclusions:

H11a: Perceived risk regulates the relationship between immersion and intention for continuous use.
H11b: Perceived risk regulates the relationship between perceived pleasure and intention for continuous use.
H12a: Perceived sociality regulates the relationship between immersion and intention for continuous use.
H12b: Perceived sociality regulates the relationship between perceived pleasure and intention for continuous use.

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4. DATA ANALYSIS

4.1 Development and Data Collection of Measurement Scales

The data collection in this study focuses on the questionnaire survey method. This process includes two stages. The first stage intends to form an initial questionnaire based on relevant research literature. To ensure the content validity of the latent and measured variables, the mature measurement scales of the relevant studies were used in the design of the questionnaire. Thereafter, these scales were revised and combined with the use of mobile devices for a reading research scenario. Perceived usefulness, satisfaction, and continuous use intention were revised based on (Yang, 2015; Zhao & Ren, 2017). Immersion was based on (Zhu, Fang & Liu, 2017). Perceived timeliness was changed from (Zhao & Ren, 2017). Information interaction quality, interpersonal interaction quality, and system interaction quality were revised according to (Wang, Yang, Li & Wang, 2017). Measurement questions were set for the 11 variables in the form of a 7-point Likert scale to indicate the degree of consent (1 = strongly disagree and 7 = strongly agree). After the initial questionnaire was drafted, 30 undergraduate students from Wuhan Business University were invited to participate in the pre-investigation. The final content and format of the formal questionnaire were modified based on the feedback and test results.

The second stage involves a formal questionnaire to clarify the cognitive and belief statuses surrounding the use of mobile devices for reading, as well as consider the coverage of the sample and quality of the survey results. Studies have identified that the younger generations are the targeted users of mobile devices for reading. This research selected a survey target of mobile device users for reading. Researchers introduced the study's purpose to the qualified participants and asked them to complete the entire questionnaire. The survey, which ran from December 2018 through February 2019, was conducted by WJX.COM, a leading online survey system in China. The questionnaire was distributed to interviewees through the WJX.COM sample service. Confirmed survey participants were given a reward. A total of 348 questionnaires were obtained. Twenty-one invalid results were eliminated, which left 327 valid responses.

4.2 Demographic Characteristics of the Sample

Table 1 illustrates the sample characteristics of the survey. Among the respondents, the proportion of female and male users was 51.07% and 48.93%, respectively. The users' age groups indicated that age was concentrated at the 21-40 age range (21-30 accounted for 44.34% and 31-40 accounted for 40.06%). Regarding education level, bachelor's degrees accounted for 83.49%, master's degrees accounted for 9.79%, and doctorates accounted for 1.22%. Regarding the occupational distribution of users, the company's white-collar workers accounted for 42.51% and technical personnel accounted for 23.55%. The income distribution of users was relatively even, with a ratio of 4001-6000 RMB to 25.38% and a ratio of 6001-8000 RMB to 22.02%.

4.3 Reliability and Validity Test

The reliability and validity of the measurement scale should be analyzed to ensure the validity of the measurement model. The measurement items in this study were from local and overseas mature scales. Therefore, the preliminary questionnaire was pre-tested, making the content of the scale reasonably clear and effective.

Reliability was tested using the composite reliability (CR) and internal consistency coefficient (Cronbach's alpha) of the latent variables. Generally, the CR value and Cronbach's alpha of the latent variable reaching 0.7 can demonstrate the reliability of the measurement model. Table 2 shows the results of the confirmatory factor analysis on the measurement scale. Moreover, Table 2 indicates that the CR values were all large and equal to 0.807. The Cronbach's alpha values were all above 0.651, thereby indicating that the measurement scale has good reliability.

The validity test includes convergence and discriminant validities, which are mainly measured by AVE. AVE is the squared difference of extraction. Generally, when AVE of all factors of the

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Table 1. Statistics of the sample's basic information

Variable	Category	Frequency	Percentage
Age	20 and below	19	5.81%
	21-30	145	44.34%
	31-40	131	40.06%
	41-50	29	8.87%
	Above 50	3	0.92%
Gender	Male	160	48.93%
	Female	167	51.07%
Education	Below high school	3	0.92%
	High school	15	4.59%
	Junior college/Bachelor's	273	83.49%
	Master's	32	9.79%
	Doctorate	4	1.22%
Occupation	Students	30	9.17%
	Civil servants	32	9.79%
	Technicians	77	23.55%
	White collar	139	42.51%
	Workers	19	5.81%
	Self-employed workers	6	1.83%
	Freelancers	24	7.34%
Monthly income	Below 2000 RMB	31	9.48%
	2001-4000 RMB	55	16.82%
	4001-6000 RMB	83	25.38%
	6001-8000 RMB	72	22.02%
	8001-10000 RMB	44	13.46%
	Above 10000 RMB	42	12.84%
How many hours per day do you use mobile devices for reading?	< 1 hour	31	9.48%
	1-2 hours	55	16.82%
	2-3 hours	83	25.38%
	> 3 hours	72	22.02%
Habit of using mobile devices for reading	Turn page with up and down	98	29.97%
	Turn page with left and right	163	49.85%
	Turn page with simulation book	66	20.18%
How long have you been using mobile devices for reading?	< 1 year	30	9.17%
	1-5 years	196	59.94%
	5-10 years	80	24.46%
	> 10 years	21	6.42%

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model is above 0.5, the convergence validity of the latent variable is considered good. When the square root of the latent variable AVE is higher than the correlation coefficient of the other related variables, the model is considered to have a good discriminant validity. If the AVE values in Table 2 are all above 0.577, then the measurement model has an ideal convergence validity. Table 3 shows that the square root of a latent variable AVE was higher than the correlation coefficient between the latent variable and other latent variables. This result indicates that the measurement model has a good discriminant validity.

5. RESEARCH RESULTS

Partial least square (PLS) is a new method of multivariate data analysis. Compared with other methods, PLS calculation results are more reliable and stable. PLS is also suitable for the analysis of small sample data. It can achieve modeling prediction, comprehensive simplification of multivariable systems, and correlation analysis between two sets of variables. It can also solve collinearity problems effectively. The purpose is to construct regression models between multiple dependent and independent variables. PLS can set the external relationship type in the structural equation flexibly according to the actual situation when constructing the model. That is, it supports the constitutive model and reflective model. This study used the Smart PLS3.0 software to analyze the model.

5.1 Path Coefficients and Hypothesis Testing

The path coefficient indicates the strength of the relationship between the independent and dependent variables. Table 4 shows the results of the path coefficient analysis of the proposed model. All hypotheses are supported.

R^2 is the variance variability explained by the dependent variable. This research used the bootstrapping repeated sampling method to select 1,000 samples to calculate the t value of the significance test. The interpretation degrees of immersion, perceived pleasure, service quality, and intention for continuous use were 0.141, 0.369, 0.353, and 0.516, respectively. The results indicated that the model has a good interpretation effect.

This study used the bootstrapping method to test the significance of the path coefficients of the structural model. Table 4 shows the results. Perceived usefulness significantly affects perceived pleasure ($\beta = 0.193$, $t = 2.964$), and significantly affects users' continuous use intention through perceived pleasure, which is consistent with the results of (Zhao & Ren, 2017). Thus, H1 is proven. Perceived interest had a significant positive effect on perceived pleasure ($\beta = 0.393$, $t = 6.185$). Hence, H3 is supported. Perceived pleasure had a significant positive effect on perceived service ($\beta = 0.595$, $t = 13.616$), thereby proving H4. Perceived pleasure had a significant positive effect on immersion ($\beta = 0.348$, $t = 4.230$). Thus, H7 is proven. Perceived pleasure had a significant positive effect on intention for continuous use ($\beta = 0.369$, $t = 5.625$). (Yang, 2015) supported this conclusion. Thus, H8 is supported. Immersion had a significant positive effect on intention for continuous use ($\beta = 0.205$, $t = 4.374$), thereby proving H9. Perceived service had a significant positive effect on intention for continuous use ($\beta = 0.207$, $t = 3.243$). Hence, H10 is proved. However, H2, H5, and H6 were not confirmed.

5.2 Regulatory Effect Test

The independent and adjusted variables were processed centrally to test the regulatory effects of the perceived risk and perceived sociality. The product terms of perceived risk, perceived sociality, perceived pleasure, and immersion were constructed; a multi-level regression analysis was performed. Perceived sociality has a regulating effect on the relationship between perceived pleasure and intention for continuous use. It has a regulating effect on the relationship between immersion and intention for continuous use.

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Table 2. Reliability and validity analyses

Variable	Measured item	Factor load	Cronbach's α	CR	AVE
Perceived usefulness	PU1	0.775	0.800	0.870	0.625
	PU2	0.797			
	PU3	0.785			
	PU4	0.806			
Perceived accessibility	PA1	0.780	0.781	0.858	0.602
	PA2	0.783			
	PA3	0.792			
	PA4	0.747			
Perceived interest	PI1	0.807	0.756	0.845	0.577
	PI2	0.769			
	PI3	0.703			
	PI4	0.757			
	PI5	0.755			
Perceived credibility	PC1	0.729	0.770	0.867	0.686
	PC2	0.905			
	PC3	0.841			
Perceived timeliness	PT1	0.801	0.761	0.862	0.675
	PT2	0.823			
	PT3	0.840			
Perceived pleasure	PP1	0.781	0.732	0.867	0.686
	PP2	0.844			
	PP3	0.794			
Immersion	IM1	0.863	0.835	0.900	0.749
	IM2	0.864			
	IM3	0.870			
Perceived service	PS1	0.835	0.752	0.858	0.669
	PS2	0.820			
	PS3	0.797			
Perceived risk	PR1	0.954	0.779	0.893	0.807
	PR2	0.839			
Perceived sociality	PSO1	0.706	0.651	0.807	0.585
	PSO2	0.853			
	PSO3	0.727			
Intention for continuous use	ICU1	0.715	0.717	0.841	0.640
	ICU2	0.863			
	ICU3	0.815			

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Table 3. Differential validity

	ICU	PC	PP	PA	PU	PS	IM	PSO	PI	PR	PT
ICU	**0.800**										
PC	0.430	**0.828**									
PP	0.621	0.523	**0.807**								
PA	0.328	0.446	0.438	**0.776**							
PU	0.338	0.405	0.475	0.432	**0.791**						
PS	0.528	0.592	0.595	0.457	0.396	**0.818**					
IM	0.415	0.094	0.344	0.232	0.125	0.212	**0.866**				
PSO	0.408	0.323	0.308	0.093	0.299	0.347	0.186	**0.765**			
PI	0.446	0.527	0.577	0.609	0.591	0.507	0.333	0.254	**0.760**		
PR	−0.104	−0.197	−0.130	0.090	−0.102	−0.117	0.172	−0.006	−0.011	**0.898**	
PT	0.508	0.526	0.563	0.591	0.439	0.559	0.255	0.218	0.552	0.030	**0.822**

Table 4. Results of the hypothesis testing

Hypothesis	Path	Path coefficient	S.E.	*t*	*p*	Whether supported or not
H1	Perceived usefulness→perceived pleasure	0.193	0.065	2.964	0.003	Yes
H2	Perceived accessibility→perceived pleasure	0.115	0.076	1.514	0.130	No
H3	Perceived interest→perceived pleasure	0.393	0.064	6.185	0.000	Yes
H4	Perceived pleasure→perceived service	0.595	0.044	13.616	0.000	Yes
H5	Perceived credibility→immersion	−0.164	0.089	1.837	0.066	No
H6	Perceived timeliness→immersion	0.145	0.082	1.781	0.075	No
H7	Perceived pleasure→immersion	0.348	0.082	4.230	0.000	Yes
H8	Perceived pleasure→intention for continuous use	0.369	0.066	5.625	0.000	Yes
H9	Immersion→intention for continuous use	0.205	0.047	4.374	0.000	Yes
H10	Perceived service→intention for continuous use	0.207	0.064	3.243	0.001	Yes

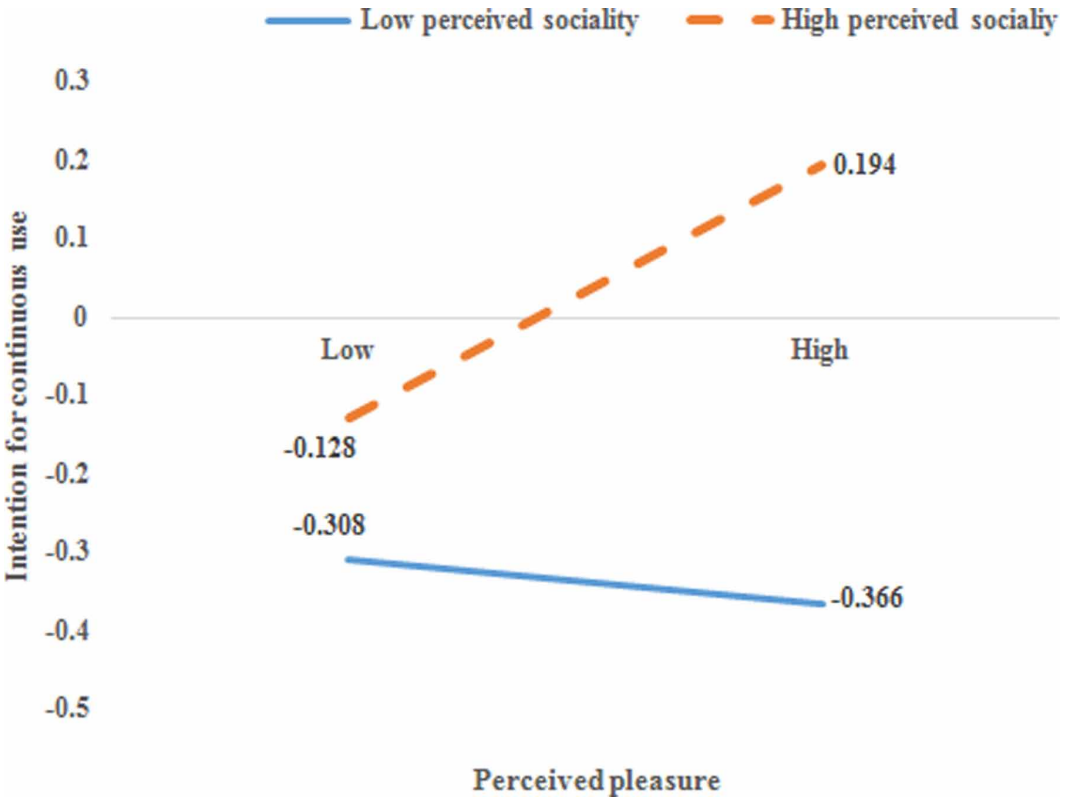
The perceived sociality regarding users of mobile devices for reading in this study is divided into high and low situations. It is convenient to clarify the role of the regulating variable. Excel was used to plot the effect of perceived pleasure on the intention for continuous use in two situations. The main, moderator, and moderation effects were 0.066, 0.185, and 0.095, respectively.

Figure 2 shows the regulating effect. The results show that when the perceived sociality of the use of mobile devices for reading users was high, the effect of perceived pleasure on intention for continuous use was weak. Hence, H12b is confirmed.

To reflect the moderating role of the use of mobile devices for reading users' perceived sociality, the average number and standard deviation of perceived sociality were calculated by taking the average plus or minus one standard deviation as the two levels of perceived sociality regarding the use of mobile devices for reading users. The main, moderator, and moderation effects are 0.584, 0.185, and -0.097, respectively. Figure 3 shows the regulating effect. When perceived sociality was high, the perceived immersion had a strong effect on the intention for continuous use. Thus, H12a is confirmed. This conclusion was also proven in (Zheng et al., 2019).

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Figure 2. Regulating effect of perceived sociality on the relationship between perceived pleasure and intention for continuous use



6. CONCLUSION

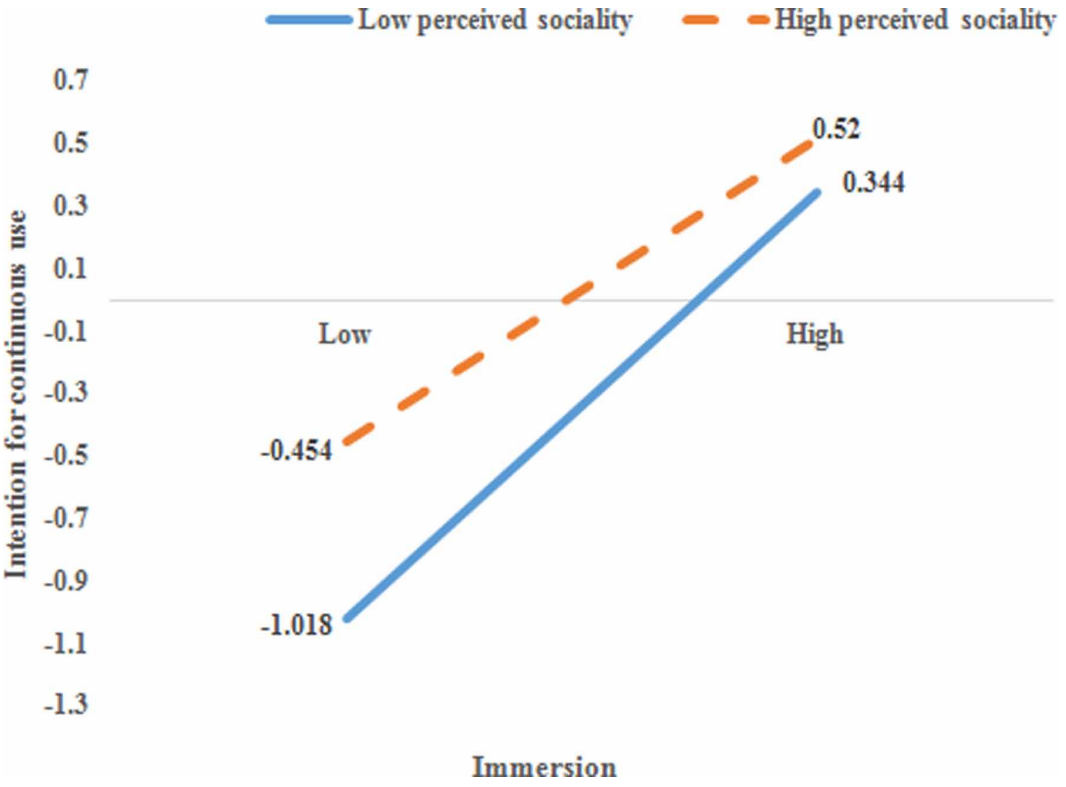
This study used the SOR model and characteristics of the use of mobile devices for reading users to analyze factors that impact the continuous use of mobile devices for reading users from the following aspects: (1) qualities of social interaction; (2) information interaction; and (3) system interaction. Consequently, a research model of the factors affecting the continuous use of mobile devices for reading users was constructed. The results present both theoretical and practical values.

The theoretical significance of this research is as follows. First, it confirms the widespread applicability of the continuous use behavior model of information systems. Perceived usefulness and perceived interest are important variables that affect users' intention for continuous use. Second, this study enriches and expands the continuous use behavior model of information systems. This research considers users' cognitive (i.e., perceived usefulness and perceived interest) factors and adds social and emotional factors like immersion. The interpretation effectiveness of the proposed model is also improved. Last, this study enriches research results regarding user behaviors and the use of mobile devices for reading. The complex effects of various variables are analyzed and verified, providing a valuable reference for understanding factors that affect continuous use intention of users of mobile devices for reading.

The practical significance of this research is as follows. This study supports service providers of mobile devices for reading in retaining existing users and increasing customer stickiness. The following aspects should be considered in terms of improving users' perceived usefulness: (1) service providers of mobile devices for reading should consider users' information needs; (2) content producers should be encouraged to create high-quality original work; and (3) providers should prioritize the improvement

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Figure 3. Regulating effect of perceived sociality on the relationship between perceived immersion and intention for continuous use



of content quality of mobile devices for reading to meet the many needs of users based on content diversification, personalization, and satisfaction. Additionally, service providers of mobile devices for reading should: (1) focus on the cultivation of users' immersion; (2) build strategic relationships with users based on their psychological needs; (3) strengthen communication with users; and (4) strive to establish emotional connections between products and users. The function of services for the use of mobile devices for reading should be improved to cultivate users' reading habits.

7. RESEARCH LIMITATIONS AND PROSPECTS

This research studied factors that affect the continuous use of mobile devices for reading from the aspects of perceived accessibility, perceived usefulness, perceived pleasure, perceived immersion, and perceived sociality. However, it has its limitations.

First, this study used a cross-sectional study of the research object based on a certain time node. To provide an improved explanation of the influential factors of the continuous use of mobile devices for reading, a set of diachronic research plans can be designed to: (1) conduct tracking research on the changes of the same batch of research objects over time; (2) investigate and compare the perceived changes of social, emotional, and cognitive factors of users in the different stages of the use of mobile devices for reading; and (3) determine the factors or change of the same factor in different stages that affect the continuous use intention of users of mobile devices for reading.

Second, although the model proposed in this study has been proven to substantially explain the continuous use intention of users of mobile devices for reading, some results may not be applicable to other countries. This study's respondents used mobile devices for reading in China. Future studies

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should expand the collection of samples to other countries, focusing on multiple cultural contexts. Future studies can also offer a comparative study of the complex effects of various model variables. These additions can improve the interpretation of the proposed model.

Last, the proposed model has systematically analyzed the occurrence mechanism that affects the use of mobile devices for reading users' continuous use intention from the three levels of SOR. A representative construct is selected in the variable measurement. This aspect inevitably disregards the impact of other variables on the continuous use intention of users of mobile devices for reading. In the future, variables should be added to the research model to offer improvements.

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Conflicts of Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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The Dark Side of Technology-Enabled Teaching:
Impact of Technostress on Student Performance

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ABSTRACT

This study validates an instrument used to measure technostress creators, technostress support mechanisms, and their negative impact on students' satisfaction and performance. A research model is developed based on the stimulus, organism, and response model to analyse the mediating effect of technostress creators and understand how technostress inhibitors influence students' satisfaction and their performance. A group of 206 students from India pursuing higher education were selected as a sample to validate this model. Technostress creators act as a mediator between technostress inhibitors and students' satisfaction and their performance, while technostress inhibitors positively influence student satisfaction and performance indirectly. Insights from this study will enable higher education institutions to identify the students who are finding technology-based education problematic and help preserve their wellbeing by following supportive strategies to reduce stress and enhance the students' active participation in technology-based education.

KEYWORDS

Student Performance, Technology-Enabled Teaching, Technostress, Technostress Inhibitors

INTRODUCTION

A crucial part of dealing with COVID-19 is to ensure that people have convenient access to services with minimum disruptions. In order to limit loss of life and spread of the coronavirus, the government of India announced a stringent lockdown on March 21, 2020. One of the many impacts of the lockdown was that it speeded up the adoption of emerging technologies in all possible fields. Industries were forced to boost both quality and delivery time of their services and this held good for the education sector as well. Technology-enabled teaching makes learning more productive and competitive, enhancing both technical skills and knowledge (Battaglino et al., 2012; Meyer & McNeal, 2011). However, acceptance of new technology requires a change in mindset of students, educationalists, policymakers, and government agencies. Many studies suggest that given proper infrastructure and skilled educators, technology-enabled teaching will boost the academic performance of students (Battaglino et al., 2012; Biagi & Loi, 2013; Genlott & Grönlund, 2016; Jena, 2015; Meyer & McNeal, 2011; Pagani et al., 2016) but many Indian educational institutions are not ready for technology-

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enabled classrooms or online classes. Overall, the quality of education will be measured not only by the quality of the faculty but also by the quality of the infrastructure for Information, Communication and Technology (ICT) and the capacity to use the technology (Biagi & Loi, 2013; De Witte & Rogge, 2014; Pagani et al., 2016; Tolani-Brown et al., 2010).

A sudden change in the teaching and learning process can affect the satisfaction of the students and their performance. The University Grants Commission (UGC), an Indian higher education regulator, has asked institutions of higher education to continue their teaching-learning process using technology-enabled teaching tools. UGC has provided guidelines to the universities that stipulate that educators should be trained in such a way that they complete about 25% of the syllabus through online teaching. It has also requested the higher educational institutions to establish virtual classrooms and video conferencing infrastructure and train all teaching staff in using the technology. UGC further suggests that institutions of higher education adopt mentor-mentee counselling through a dedicated platform to provide students with appropriate guidance (UGC, 2020). However, the guidelines provided by UGC focus on the perspectives of institutions for higher education and educators and fail to address the students' acceptance of technology-enabled teaching and/or the issues they face related to technology-enabled teaching. Many Indian institutions have accelerated the adoption of technologies to deliver classes, but digital literacy is almost non-existent among over 90% of India's population (ICT Academy, 2020). Issues such as slow Internet connectivity, lack of telecom infrastructure, online system affordability, availability of personal systems and software will make digital teaching more complex. These reasons could create stress for students of generation Z, even if they are highly exposed to technology (Jena, 2015; Jena & Mahanti, 2014; Shu et al., 2011; X. Wang et al., 2020).

Earlier studies have focused on technostress creators, the support mechanisms to reduce technostress, and its negative impact on outcomes (Shu et al., 2011; Tarafdar et al., 2011, 2015; Wang et al., 2005). All these studies were done in developed countries and primarily focused on the organisational context. However, there is lack of evidence to measure levels of technostress based on technostress creators, support mechanisms and technostress impacts from the perspective of students completing their higher education in developing countries, even though a few studies have been done from the academicians' perspective (Jena, 2015; Jena & Mahanti, 2014). Therefore, this study aims to fill this gap by validating an instrument to measure technostress creators, technostress support mechanisms and the negative impact of technostress on students' satisfaction and performance. This study aims to provide better insights to identify students who find technology-enabled teaching problematic and, at the same time, help the higher education institutions preserve their wellbeing by following supportive strategies that reduce stress and enhance their active participation in technology-enabled teaching.

LITERATURE REVIEW

Technostress

Technostress has been defined as an incapability of the human mind to cope with new technological interventions (Ayyagari et al., 2011). This incapability to adapt to a new technology starts with an uneasy feeling that arises due to various reasons, such as limited knowledge about technical functions, low technical literacy, and limited technical support; and leads to anxiety and stress. This anxiety, if not taken care of, will lead to various psychological and physical problems. Arnetz & Wiholm (1997) used this word and tried to explain the psychological and mental state of people struggling to adopt technology in their work. Johansson-Hidén et al. (2003) described this term as ICT stressor, which affects a person's mental health and their productivity. ICT has been introduced in organisations to reduce human anxiety and tension; however, it has sometimes been seen to create insecurity and uncertainty, leading to a feeling of inadequacy and stress (Arnetz & Wiholm, 1997; Ragu-Nathan et al., 2008; Tarafdar et al., 2010, 2015). End-user productivity is compromised by continuously evolving

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technology, applications, and techniques which, in turn, lead to overload, fear of becoming obsolete, and dissatisfaction with work (Ragu-Nathan et al., 2008; Sainfort, 1990; Tarafdar et al., 2010, 2015). Research has discussed various forms of technostress, known as techno-anxiety (Lee et al., 2016; Salanova et al., 2013; Sellberg & Susi, 2014), techno-addiction (Dhir et al., 2018; Rozgonjuk et al., 2018) and techno-strain (Salanova et al., 2013).

Technostress Inhibitor

Organisations need to develop well-built technology-coping strategies to adapt to ICT changes, otherwise these stress creators will lead to technostress (Christian et al., 2020; Ragu-Nathan et al., 2008; Tarafdar et al., 2015). These coping mechanisms usually consist of ICT training and technical infrastructure to integrate these new technologies smoothly into the system. These supportive coping strategies are termed as technostress inhibitors (Ragu-Nathan et al., 2008; Tarafdar et al., 2014, 2015). To counter the negative effect of this fast-changing technological world, these inhibitors have to be used strategically. The literature suggests three technostress inhibitors, namely literacy facilitation, technical support provision, and involvement facilitation (Fuglseth & Sørebø, 2014; Li et al., 2020; Tarafdar et al., 2019). Technical support provisions are described as knowledge and support infrastructure available in the organisation to handhold the people at the time of scarcity of technical knowledge. This support can reduce the negative after-effects associated with ICT usage. This technical assistance helps to familiarize the students with new technology and resolve their problems and challenges (Chandra et al., 2019; Srivastava et al., 2015; Tarafdar et al., 2011). An organisation can extend literacy facilitation by providing different employees with technical development programs to improve their technical skills and knowledge (Fuglseth & Sørebø, 2014; Safahieh & Asemi, 2010). To avoid further delay in the process, an organisation should emphasise and try to inculcate a culture of teamwork (Shu et al., 2011; Tu et al., 2005; Wang et al., 2005). Tu et al. (2005) conclude that when an organisation provides a high level of technical assistance to its employees, it will benefit from increased productivity. Moreover, a regular technology development program would encourage employees to use more technology (Shu et al., 2011). The level of technostress can be lowered to a great extent if the employee perceives that an organisation is giving sufficient administrative assistance (Burke, 2009; Koo & Wati, 2011). The problem of technostress was reported mostly where employees were forced to use technology without their consultation and training. (Kupersmith, 1992; Liu, 2012; Poole & Denny, 2001; Tseng, 2015; Vyhmeister et al., 2006). Involvement facilitation is described as the extent to which an individual is involved in the process of planning and implementing the technology. With this involvement, the end-user would feel well informed about future actions like why, how, when, and the potential effect of new technology implementation.

Al-Qallaf (2006) found that an absence of a regular technical skill development program and sufficient training; lack of network infrastructure and involvement at the time of planning and implementing the ICT; an inadequate workforce for assisting people at the time of a breakdown; and an ever- increasing demand, and information overload were some of the many reasons leading to technostress. When literature focus is analysed, lack of formal training and user acceptance are seen to rank high as factors leading to technostress (Al-Fudail & Mellar, 2008; Buarki et al., 2011; Jena, 2015; Kupersmith, 1992; Qi, 2019; Tarafdar et al., 2010, 2019).

In general terms, extant studies have shown that technostress inhibitors may minimise the negative effects of technostress creators on the work output of people. Simple access to help desk and technical support to fix end-user encounter issues in their work, for example, may relieve stressed feelings and boost performance (Jena, 2015; Ragu-Nathan et al., 2008; Tarafdar et al., 2011). Also, the involvement of university teachers in the planning, implementation, assessment, and refinement phases of ICT integration in higher education may diminish their technostress (Brooks & Califf, 2017; Li et al., 2020). Informed by previous studies on technostress inhibitors and creators, technostress inhibitors can boost end-user competence in the use of ICT in their work, improve their work engagement and ultimately increase their work output.

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Technostress Creators

The survival of any organisation is dependent on its existence on various digital platforms, such as web and mobile applications, social media, and other collaborative applications to manage different stakeholders (Ollo-López & Aramendía-Muneta, 2012; Yunis et al., 2012). Various research papers talk about different platforms and how their excessive usage and dependence lead to technostress (Hung et al., 2015; Lee et al., 2016; Ragu-Nathan et al., 2008; Salo et al., 2019).

In organisational research, the job performance of workers is a significant factor that defines whether individual companies outperform others. The value of job performance also extends to higher education environments where students, funding support, and reputations have become increasingly competitive among universities worldwide (Wæraas & Solbakk, 2009). The deployment of organisational resources for the better performance of end-users and their well-being can be guided by knowing factors influencing university work performance. ICT is increasingly becoming an integral part of institutions of higher education and intends to improve teaching and learning. Nevertheless, for university teachers, students, the rapid iterations and advances of ICT can also cause technostress and make them feel incapable, frustrated, and stressed, thus negatively affecting their work efficiency (Brooks & Califf, 2017; Christian et al., 2020; Li et al., 2020).

Technostress concerning students in different educational institutions is an understudied subject. Individual productivity has been hampered by the proliferation of ICT and ICT-developed platforms at different levels. Each individual has different competency levels of ICT usage, a different mindset, and orientation towards ICT, and different experiences with such platforms but with the combination of these, there is the necessity of a successful implementation of ICT in the current situation (Ayyagari et al., 2011; Ragu-Nathan et al., 2008; Srivastava et al., 2015; Tarafdar et al., 2015).

Individual productivity is affected by different stressors, which are defined as stress creators in different research papers (Chandra et al., 2019; Hwang & Cha, 2018; Marchiori et al., 2019; Ragu-Nathan et al., 2008). The factors that cause technostress can be defined as techno-overload, techno-complexity, techno-uncertainty, techno-invasion, and techno-insecurity. Previous studies have mostly talked about the first three factors in the list, i.e., techno-overload, techno-complexity, and techno-uncertainty (Alam, 2016; Ragu-Nathan et al., 2008) whereas Giorgi et al. (2015) and Tarafdar et al. (2010) have discussed techno-invasion and techno-insecurity.

Tarafdar et al. (2011) portrayed the relationship between technostress creators, technostress inhibitors, and their outcome using factors such as techno-overload, techno-invasion, techno-complexity, techno-insecurity, and techno-uncertainty as the components of technostress.

Techno-overload is a scenario where employees are expected to work for longer duration and at a faster speed than before the implementation of ICT due to increased work demand (Tarafdar et al., 2011, 2019). Olaniyi et al. (2014) concludes that techno-overload can result in employees experiencing fatigue and other health problems. Techno-overload contributes to a large number of cumulative trauma disorders that affect the hands, wrists, elbows, arms, shoulders, lower back and cervical spine areas. Studies conclude that techno-overload can cause the emotional exhaustion of the end-user (Christian et al., 2020; Tarafdar et al., 2014, 2015).

Techno-invasions refer to a situation in which organizations expect their employees to be available at any time and every time and be always connected with their work-related affairs (Ragu-Nathan et al., 2008; Tarafdar et al., 2014). Together with techno-addiction, techno-invasion entails that work-related tasks may spill into the worker's private life, endangering their work-life balance (Brivio et al., 2018). The comfort offered by technology also forces workers to conduct work in non-working hours. Therefore, the distinction between work and family becomes blurred, which creates tension between work and family, thus, resulting in a vicious circle. At the same time, people have feelings of being bound to technology; technology disrupts the time and space of workers, making them feel nervous about work. For example, learning new technology or working overtime on weekends can lead to employees' shortage of investment in family time, which causes work-family conflict and

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results in increased levels of job stress (Brivio et al., 2018; Jena & Mahanti, 2014; Tarafdar et al., 2011, 2014, 2019).

Techno-complexities are described as those moments when an employee feels incapable and low on skills because of the ever-evolving technologies and techniques (Tarafdar et al., 2010, 2014). This situation is created by technological complexities, for instance, the introduction of new software or new platforms. These techno-complexities mean that employees have to constantly reskill and upskill in line with the evolving technologies, which leads to immense pressure to cope with challenges associated with ICT implementation (Brooks & Califf, 2017). Teaching and learning processes with the use of ICT become more complicated than conventional methods in face-to-face classes. Various complicated procedures are present in the integration of ICT in the teaching and learning process. When the end-users fail to understand the critical aspects of ICT in the teaching and learning process, they are more averse to using it. Since the new ICTs are multifaceted and require tremendous effort to understand, in the long run, these kinds of complexity create stress and affect end-user performance (Al-Fudail & Mellar, 2008; Hatlevik & Hatlevik, 2018; Jena, 2015; Li et al., 2020).

Techno-insecurity is described as a situation where employees are always under fear and insecurity of job loss and of being replaced by someone who has a better understanding of technology. Li et al. (2020) explain techno-insecurity in the educational setting. Techno-insecurity is a situation where teachers feel insecure about the presence of ICT because they fear that it could make them redundant. Recent studies conclude that students sometimes take their life because of lack of education due to lack of access to technology (Balachandran et al., 2020; Lathabhavan & Griffiths, 2020; Mamun et al., 2020). Techno-insecurity is motivated by basic things, such as lack of confidence and anxiety in using ICTs in the teaching-learning process, and poor experience in dealing with operational issues related with ICT such as lack of access to ICT facilities and poor connectivity (Shapka & Ferrari, 2003; Vannatta & Fordham, 2004; Wang et al., 2008).

Techno-uncertainty refers to those situations that are not predictable. This unpredictability and continuous change leads to an unstable and unsettled mindset (Ragu-Nathan et al., 2008; Tarafdar et al., 2010, 2015). Technology is changing the dynamics of education, especially the relationship between teachers and students. Many higher educational institutions have already adopted ICT in the teaching-learning process, and there are many others that are in the process of doing so. However, the continuous evolution of technology poses challenges for the educational sector. Some of these challenges include frequent updates and upgrades in ICT, increased cost in operationalizing and maintaining ICTs, and replacing one ICT with another. All these add to techno-uncertainty among teachers and students (Al-Fudail & Mellar, 2008; Burke, 2009; Jena, 2015; Jena & Mahanti, 2014).

End-User Satisfaction

ICT implementation can give rise to several issues, making it challenging to understand and use features. A user always looks for easy to use technology and friendly functional features to access rich content in their desired format (Kanter & Moss, 1985). New technology sometimes creates fear in the mind of employees and limits their capabilities and hinders innovation, experimentation, and risk-taking (Chua et al., 1999; Ragu-Nathan et al., 2008; Tarafdar et al., 2015).

End-user satisfaction refers to the positive response of employees towards the implementation of new software and applications (Domínguez-Escrig et al., 2018; Wixom & Todd, 2005). The use of new technology or software should be effortless (Doll & Torkzadeh, 1988). End-user satisfaction increases if the information provided by the system is accurate, relevant, and timely. This will enable greater comfort with the technology, leading to a stress-free user experience, which in turn leads to an increase in user productivity and innovation (Ragu-Nathan et al., 2008; Tarafdar et al., 2014). There is also improved decision-making by both the employee and the management (DeLone & McLean, 2016; Tarafdar et al., 2011). End-user satisfaction is the degree up to which employees accept new technology in the organisation. Sometimes it is measured quantitatively with the help of output of the technology-enabled work or by improvement in their task efficiency, productivity, and innovation

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(Ragu-Nathan et al., 2008; Tarafdar et al., 2014, 2015, 2019). User involvement can be improved by giving proper training on technological changes; in this context, the technostress inhibitors help the users to understand the real benefits of ICT used in the organisation. Also, end-users will be happy to use the ICT in the workplace if the inhibitors are appropriately addressed before the implementation of ICTs (Fuglseth & Sørebø, 2014; Tarafdar et al., 2011).

End-User Performance

The implementation of ICT in an organisation has changed the nature of work (Saltari et al., 2013). These changes in technology are real-time, complex, incremental, and require the user to use different platforms to capture and process information. The user has to work on the different applications simultaneously, which demands them to be more cautious during the final presentation of the data. ICT users deal with a plethora of information, face constant connectivity issues, and engage in multitasking. Users have to put in continuous efforts to master the frequently changing technology (Ragu-Nathan et al., 2008; Tarafdar et al., 2014, 2015, 2019). Technostress leads to several behavioural issues such as anxiety, panic attacks, procrastination, and insecurity of being replaced by someone who is better at handling technology (Ragu-Nathan et al., 2008; Tarafdar et al., 2014, 2015, 2019; Tu et al., 2005; K. Wang et al., 2005). Studies have verified that the presence of high-level stressors increases the chance of task-specific mistakes and accidents (Kahn & Byosiere, 1992). All these issues affect job satisfaction and the work commitment level of employees. It also creates a negative impact on the creativity and productivity of employees (Tarafdar et al., 2011, 2015).

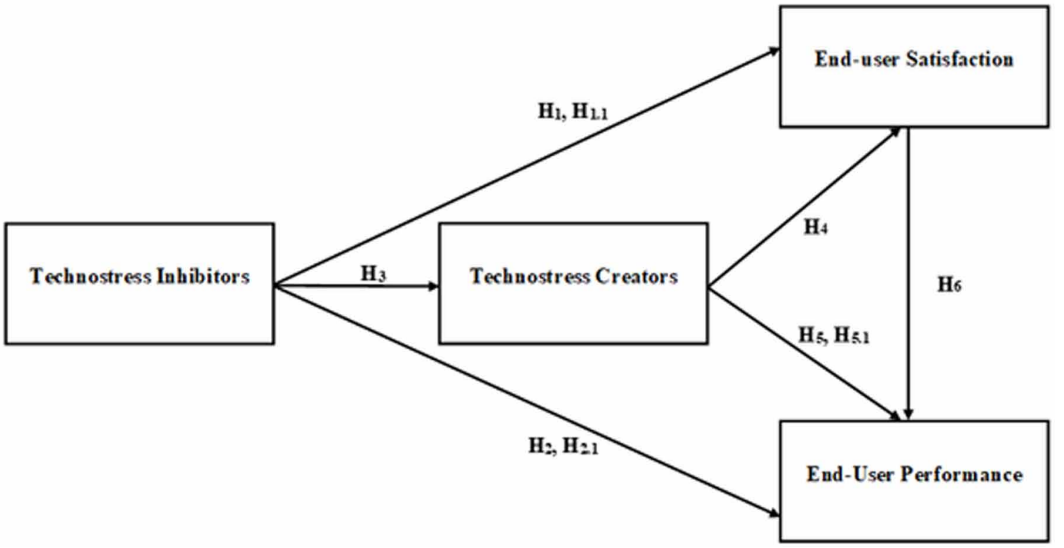
If organisation inhibitors are worked on effectively and embedded in the system properly, then the impact of technostress creators can be reduced on technology-led innovation and performance. If employees are given training before implementing any new application, their self-belief about technology usage will become positive, and stress creators will have a relatively lower negative impact on performance (Ragu-Nathan et al., 2008; Tarafdar et al., 2010, 2015, 2019). There is a negative relationship between individual job outcomes and technostress. Past research on stress and coping shows that there is an inverted U-shaped relationship between stress and job outcome. The same study conducted in the context of technostress indicates that there is a possibility of an inverted U-shaped relationship between technostress creators and ICT-enabled job outcome (Abualrub & Al-Zaru, 2008; Chandra et al., 2019; Srivastava et al., 2015). Studies show that a moderate level of technostress creators in the organisation leads to higher job outcomes.

Conceptual Model and Hypothesis Development

Many environmental psychology and service marketing studies have used the S-O-R model (Bitner, 1992; Buckley, 1991; Cao & Sun, 2018; Choi & Kandampully, 2019; Luqman et al., 2017; Noble, 1966; Platts, 1972) to explain that the various aspects of the physical evidence act as stimuli (S) which affect the internal states of individuals (O), and which in turn, influence the behavioural responses (R). The model illustrates that an individual's inward states are reinforced by the "Stimuli" found in external aspects. The word "Organism" has been used to refer to the inner states of an individual's perception, emotions, and thought. Previous studies have viewed those constructs to be both positive and negative. Finally, individuals make their final choice and accordingly choose their behavioural "Response." Based on the literature discussions and the S-O-R model, similar theoretical links can be used for the present study. In this study, the technostress inhibitors are considered the stimuli that motivate students to use technology-enabled teaching, and which reduce the behavioural response otherwise called technostress. Finally, reduced technostress positively influences the students' perception and emotions about ICT usage in education, which leads to increased user satisfaction and performance. Moreover, from this model, it can be concluded that technostress creators act as a mediator between technostress inhibitors and students' outcomes in terms of end-user satisfaction and performance. Based on the above discussions and theoretical model, we, therefore, frame the following hypotheses (A conceptual model is shown in Figure 1).

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Figure 1. Conceptual Framework



H_1: Technostress inhibitors positively influence students' satisfaction
$H_{1.1}$: Technostress creators mediate the relationship between technostress inhibitors and students' satisfaction
H_2: Technostress inhibitors positively influence students' performance
$H_{2.1}$: Technostress creators mediate the relationship between technostress inhibitors and students' performance
H_3: Technostress inhibitors negatively influence technostress creators
H_4: Technostress creators negatively influence students' satisfaction
H_5: Technostress creators negatively influence students' performance
$H_{5.1}$: Students' satisfaction mediates the relationship between technostress creators and students' performance
H_6: Students' satisfaction positively influences students' performance

METHODS

Sampling

A quantitative methodology was adopted, and an online survey was designed to collect the responses from university/college students from Bangalore, India. The higher education students participating in this study came from four private universities in Bangalore. These universities have implemented a variety of ICT in their classrooms and follow blended learning practices. However, due to the COVID-19 pandemic, as per the UGC directions, these universities adjusted their curricula to fit entirely on online-based teaching. As a result, many higher education students needed to make changes to their learning practice to adapt to ICT-enhanced pedagogy. Data was collected from March 2020 to June 2020. Two hundred and fifteen students pursuing higher education were selected for this study, and the random sampling method was used. Incomplete and invalid samples were removed leaving 206 samples for the study. The data were collected through an online survey. The sample was taken from students from India pursuing higher education and consisted of undergraduate students (10%), and postgraduate students (90%). The ages of respondents ranged from 18 to 29 years, with an average of 22.78 years. Fifty-five percent of the participants were male, and 45% of the samples were female.

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Before conducting the survey, the volunteers were provided with a survey form which clarified the aim of the research. They were also assured that the data so collected would remain confidential and anonymous. IBM SPSS 25 and SmartPLS 3 software packages were used to analyse data.

Measures

This research consists of four sections. The first section of the instrument consists of a technostress construct. Technostress creators are used to measure stress experienced by the students when experiencing technology-enabled teaching in higher education. The technostress creator is measured by five subdimensions, namely Techno-overload, Techno-invasion, Techno-complexity, Techno-insecurity, and Techno-uncertainty (Tarafdar et al., 2010). This technostress creator scale consists of 23 items.

The second section includes the items related to technostress inhibitors, adapted from Ragu-Nathan et al. (2008). Technostress Inhibitors were intended to ask respondents' opinions about supportive mechanisms implemented by the institutions of higher education to reduce the negative effects of technostress. These inhibitors were measured by three subdimensions: Literacy facilitation, Technical support provision, and Involvement facilitation, on a scale consisting of 13 items.

The third section of the instrument included items related to end-user satisfaction and performance. The end-user performance construct consisted of 7 items and aimed to measure students' outcomes, quality, and innovation. The end-user satisfaction was intended to measure the students' satisfaction towards the technology-enabled teaching content and the accuracy, ease of use, and timeliness of the information; it consisted of 12 items, adapted from Tarafdar et al. (2010), and initially used by Doll & Torkzadeh (1988). All the items were assessed using a 7-point Likert scale ranging from "1 = extremely strongly disagree" to "7= extremely strongly agree". The individual constructs and related measures are presented in Appendix I. The last section of the instrument included the demographic details of the students, such as age, gender, educational qualification, and type of institution. The validity and reliability of the scales are discussed in the results section.

RESULTS

Measurement Model

The underlying conceptual model was assessed using the Partial Least Squares- Structural Equation Modeling (PLS-SEM) approach using SmartPLS. The reason for using PLS is that it is a flexible modeling approach to SEM with no data distribution assumptions. PLS-SEM is good to use when the sample size is small, the distribution of data is skewed and there is a limited availability of theories to applications (Vinzi et al., 2010; Wong, 2010). Since the sample size was small, and the data was non-normally distributed, SmartPLS 3 software was used to analyse the measurement model and the structural model.

This section explains the validity of the four measurement models and internal consistency reliability. Cronbach's alpha (α) and composite reliability (CR) were the main estimates for measuring internal consistency reliability (Hair et al., 2014). The validity of the measurement models was estimated by checking the convergent and discriminant validity. Convergent validity checks how well the items measure a related construct. In PLS-SEM, Average variance extracted (AVE) and factor loadings of the items (outer loadings) are used to evaluate the convergent validity of the models.

Table 1, 2, 3 and 4 show that the AVE for each construct is above 0.5, which means that more than half of the variances observed in the dimensions are accounted for in the hypothesised constructs and exceed the threshold value of 0.5 or above (Hair et al., 2014), and it reveals the convergent validity (Fornell & Larcker, 1981). Convergent validity is also supported by the composite reliability (CR) of each measure, which is higher than the threshold value of 0.7 or above. Appendix A shows that

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Table 1. Technostress Inhibitors Reliability and Validity Measures

Constructs	α	CR	AVE	1	2	3
1. Involvement facilitation	0.853	0.852	0.594	**0.771**		
2. Literacy facilitation	0.875	0.872	0.580	0.584**	**0.762**	
3. Technical support provision	0.845	0.845	0.578	0.750**	0.761**	**0.760**

Note: **p<0.01; Diagonals value the square root of the AVE

Table 2. Technostress Creators Reliability and Validity Measures

Constructs	α	CR	AVE	1	2	3	4	5
1. Techno-complexity	0.827	0.877	0.592	**0.769**				
2. Techno-insecurity	0.835	0.883	0.602	0.452**	**0.776**			
3. Techno-invasion	0.802	0.87	0.626	0.310**	0.352**	**0.791**		
4. Techno-overload	0.857	0.897	0.636	0.336**	0.301**	0.343**	**0.797**	
5. Techno-uncertainty	0.802	0.868	0.623	0.307**	0.214*	0.044	0.271*	**0.789**

Note: **p<0.01; *p<0.05; Diagonals value the square root of the AVE

Table 3. End-user Satisfaction Reliability and Validity Measures

Constructs	α	CR	AVE	1	2	3	4	5
1. Accuracy	0.779	0.780	0.639	**0.799**				
2. Content	0.860	0.859	0.605	0.558**	**0.778**			
3. Ease of use	0.810	0.810	0.681	0.631**	0.494**	**0.825**		
4. Output	0.765	0.765	0.619	0.750**	0.617**	0.623**	**0.787**	
5. Timeliness	0.764	0.768	0.624	0.734**	0.461**	0.744**	0.639**	**0.790**

Note: **p<0.01; Diagonals value the square root of the AVE

Table 4. End-User Performance Reliability and Validity Measures

Constructs	α	CR	AVE	1	2
1. ICT-enabled innovation	0.862	0.862	0.675	**0.821**	
2. ICT-enabled productivity	0.859	0.858	0.603	0.768**	**0.777**

Note: **p<0.01; Diagonals value the square root of the AVE

outer loading of each item is above 0.5 for all research measures and all the items are significant at 1%, and so we retain all the items.

According to Hair et al. (2014), the high-reliability value (CR >0.95) suggests that there are redundant items in the research scale; all our CR values of the constructs are less than 0.95, which shows that there are no redundant items in the research. Cronbach's alpha for all the research constructs is higher than the acceptable threshold of 0.7, which means that there are no reliability issues in this research (Zikmund & Babin, 2010). Discriminant validity checks how well a construct varies from the other constructs in the measurement model, and it is assessed using the Fornell-Larcker criterion

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Table 5. Differences between level of technostress based on demographical variables

Technostress Creator	Male (n=113)		Female (n=93)		t value (p value)	UG (n=21)		PG (n=185)		t value (p value)
	Mean	SD	Mean	SD		Mean	SD	Mean	SD	
Techno-overload	2.38	0.89	2.23	0.83	1.196 (0.233)	2.57	1.11	2.28	0.83	1.463 (0.145)
Techno-invasion	2.18	0.80	2.19	0.92	-0.056 (0.955)	1.86	0.88	2.22	0.85	-1.862 (0.064)
Techno-complexity	3.06	1.02	2.98	1.00	0.963 (0.565)	3.19	1.18	3.00	0.99	0.798 (0.426)
Techno-insecurity	2.42	0.83	2.35	0.76	0.346 (0.559)	2.31	0.93	2.39	0.78	-0.437 (0.663)
Techno-uncertainty	2.54	0.85	2.68	0.95	-1.100 (0.273)	2.87	0.88	2.58	0.90	1.413 (0.159)

(Fornell & Larcker, 1981). Tables 1, 2, 3 and 4 show that the square root of AVE is higher than inter-construct correlations, which support the discriminant validity of the constructs (Hu & Bentler, 1999) and shows that each construct is unique in this research.

Before testing the hypothesis, the level of technostress was assessed against the demographic variables, such as gender and qualification. An independent sample t-test was used to find the difference in the level of technostress experienced by the students based on their gender and qualification. Table 5 show that male and female students have the same level of technostress. In terms of educational qualification, undergraduate and postgraduate students have the same level of technostress. Among those five technostress components, techno-complexity was the highest, and techno-invasion the lowest, level of technostress experienced by the students.

Structural Modelling and Model Fit Statistics

The path coefficients, T values were obtained by bootstrapping generated by 5000 samples to get a reasonable estimation of standard error. The structural model, along with path coefficients (β), factor loadings, and R square values, are presented in Figure 2. As seen in the figure, all the beta values are significant at the 5% level and above 0.5, showing the significance and relevance of the hypothesis links. Variance inflation factor (VIF) values are used to assess the collinearity issues. Table 6 shows that VIF values are less than 3, which is within the accepted range recommended by Hair et al. (2014) and thus, we can conclude that there are no multicollinearity issues in the research data.

Table 7 shows the model fit statistics summary. The R-square (R^2) is used to measure the proportion of the variance in the dependent variable that is predictable from the independent variable. Hair et al. (2014) suggest a cut-off range for predictive accuracy. An R^2 value between 0.330 to 0.670 means that the predictive accuracy is moderate, and above 0.670 means it is substantial. All three R^2 values range from 0.330 to 0.447, suggesting that they are acceptable; and all the constructs are moderately predicted by the endogenous constructs.

In SmartPLS, the Blindfolding method is used to assess predictive relevance (Q^2). Predictive relevance relates to the "accurate prediction of the data points of indicators in reflective measurement models of endogenous constructs and endogenous single-item constructs". Q^2 values should ideally be greater than zero to have predictive relevance. Hair et al. (2014) recommends a cut-off range for predictive relevance. A Q2 value between 0.25 to 0.35 means that predictive relevance is moderate and a value above 0.35 points to a strong predictive relevance. All three Q^2 values range from 0.219 to 0.244, suggesting that they are acceptable, and the present model has moderate predictive relevance for the endogenous constructs.

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Table 6. Collinearity Statistics

Constructs	VIF
Accuracy	1.83
Content	1.37
Ease of use	1.51
ICT-enabled innovation	1.63
ICT-enabled productivity	1.63
Involvement facilitation	1.55
Literacy facilitation	1.63
Output	1.64
Technical support provision	1.90
Techno-complexity	1.43
Techno-insecurity	1.52
Techno-invasion	1.57
Techno-overload	1.48
Techno-uncertainty	1.66
Timeliness	1.66

Note: Accepted cut-off range (VIF < 3)

The overall model fit is assessed using Goodness-of-Fit (GoF). Hair et al. (2014) defines GoF as "how well the specified model reproduces the observed covariance matrix among the indicator items". Tenenhaus et al. (2005) recommends a formula to calculate GoF, which is "the geometric mean of the average communality and average R^2 for an endogenous construct." Wetzels et al. (2009) suggest the cut-off range for accepted GoF ranges As medium between 0.25 to 0.36 is and large for above 0.36. In the present study, the three dependent variables and corresponding GoF values are above 0.36, which testifies to the perfect fit of the model.

Direct Effect

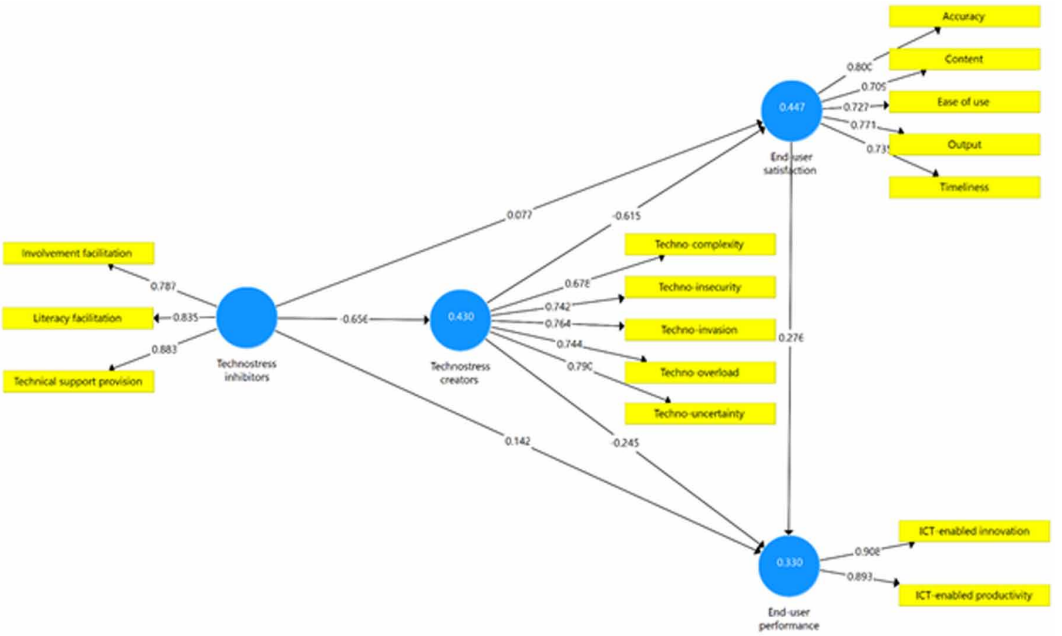
Table 7 shows the direct path relationships. Technostress inhibitors have a significant direct and negative impact on technostress creators (β=-0.656; t=16.945; p<0.01), thus, H_3 is supported, and technostress inhibitors explain 43% of the variation in the technostress creators. Technostress inhibitors do not have any direct impact on end-user satisfaction (β=0.077; t=1.044; p=0.297) and end-user performance (β=0.142; t=1.655; p=0.098); and thus, H_1 and H_2 are not supported. At the same time, the technostress creators have a direct and negative impact on end-user satisfaction (β=-0.245; t=2.412; p<0.05) and end-user performance (β=-0.615; t=8.92; p<0.01); thus, H_4 and H_5 are supported. The end-user performance is directly, significantly, and positively influenced by the end-user satisfaction (β=0.276; t=2.559; p<0.05), and hence, H_6 is supported.

Indirect Effects

Table 8 shows the indirect path relationships. The technostress inhibitors indirectly affect end-user satisfaction and performance. The end-user is significantly affected by technostress inhibitors via technostress creators. The direct effect between technostress inhibitors and end-user satisfaction is not significant. However, the indirect effect is positive and significant (β=0.404; t=7.205; p<0.01). Hence, the technostress creators act as a full mediator between technostress inhibitors and end-user

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Figure 2. Structural Model



satisfaction. Altogether, indirect and direct effects account for 45% of the variation in end-user satisfaction, we can conclude that H1.1 is supported. Technostress inhibitors influence end-user performance via three different indirect relations; 1. Technostress inhibitors → Technostress creators → End-user performance (β=0.161; t=2.345; p<0.05); 2. Technostress inhibitors → End-user satisfaction → End-user performance (β=0.021; t=0.917; p=0.359); and 3. Technostress inhibitors → Technostress creators → End-user satisfaction → End-user performance (β=0.112; t=2.244; p<0.05). The total indirect effect between technostress inhibitors and end-user performance is significant and direct effect between technostress inhibitors, and not significant for end-user performance. Hence, it can be concluded that technostress creators act as a full mediator between technostress inhibitors and end-user performance, and here, end-user satisfaction does not act as a mediator between technostress inhibitors and end-user satisfaction, thus, H2.1 is supported. Since Technostress inhibitors → End-user satisfaction → End-user performance path is not significant, it implies that technostress creators indirectly influence the end-user performance via end-user satisfaction (β=-0.17; t=2.305; p<0.05),

Table 7. Direct Effect and Model Fit statistics

Direct Paths	β	T-value (p-Value)	R^2	Q^2	GoF
End-user satisfaction → End-user performance	0.276	2.559 (0.011*)	0.330	0.244	0.517
Technostress creators → End-user performance	-0.245	2.412 (0.016*)			
Technostress inhibitors → End-user performance	0.142	1.655 (-0.098)			
Technostress inhibitors → Technostress creators	-0.656	16.945 (0.000*)	0.430	0.219	0.484
Technostress creators → End-user satisfaction	-0.615	8.92 (0.000*)	0.447	0.227	0.501
Technostress inhibitors → End-user satisfaction	0.077	1.044 (-0.297)			

Note: **p<0.01; *p<0.05

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Table 8. Indirect Effect

Indirect paths	β	T-value (p-Value)
Technostress creators → End-user satisfaction → End-user performance	-0.170	2.305 (0.021*)
Technostress inhibitors→ End-user satisfaction → End-user performance	0.021	0.917 (-0.359)
Technostress inhibitors→ Technostress creators → End-user performance	0.161	2.345 (0.019*)
Technostress inhibitors→ Technostress creators → End-user satisfaction → End-user performance	0.112	2.244 (0.025*)
Technostress inhibitors→ Technostress creators → End-user satisfaction	0.404	7.205 (0.000**)

Note: **p<0.01; *p<0.05

and so $H_{5.1}$ is supported. In this case, both the direct and indirect relationship between technostress creators and end-user performance are significant, and end-user satisfaction acts as a partial mediator between technostress creators and end-user performance. Altogether, indirect and direct effects account for 33% of the variation in the end-user performance.

DISCUSSION AND IMPLICATIONS

This study measures the levels of technostress based on technostress creators and the impact of technostress on students' satisfaction and performance. It also identifies the different support mechanisms (technostress inhibitors) used by the institutions for higher education to reduce the technostress and how it influences students' satisfaction and their performance, which have constituted a less understood view on ICT use in Indian institutions of higher education to date, with students as end-users. Prior studies have acknowledged that technology-enabled teaching has a significant and positive impact on students' performance and satisfaction; however, it is also observed that technology implementation in higher education needs to be done only after getting user acceptance. In many developed countries, such institutes follow blended learning methods by using ICTs along with their traditional teaching methods. Due to the COVID-19 outbreak, they have been forced to use ICT-enabled teaching, which has however failed to receive end-user acceptance (students), thus, directly impacting end-user satisfaction. In this view, findings from previous research have suggested both negative and positive outcomes of technology-enabled teaching.

From this study, it is evident that all three components of technostress inhibitors are significant; however, the technical support provision is the most critical component of technostress inhibitors. The performance of the end-user help desk in terms of accessibility, knowledge, and responsiveness in addressing students' queries related to technology-enabled education helps students to reduce their technostress. When it comes to components of technostress creators, techno-uncertainty and techno-invasion are the most critical components of technostress creators; however, the other three components are also considered as relatively important. Frequent changes and updates in computer software and hardware give techno-uncertainty stress. Since technology-enabled education is new to most students in India, they have to spend more time in mastering the new ICT system, which affects their personal time. Accuracy of the system and output quality are the most crucial components of end-user satisfaction, and innovation related to ICT is an essential component of end-user performance.

The present study conceptualises and empirically validates the relationship between technostress inhibitors, technostress creators, and students' learning outcomes. Findings from our study show that technostress inhibitors do not influence students' satisfaction and performance directly. Technostress creators act as mediators between technostress inhibitors and students' satisfaction and performance; and technostress inhibitors indirectly and positively influence students' satisfaction and performance, in contradiction with Ragu-Nathan et al. (2008). Next, technostress inhibitors negatively impact

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technostress creators, which directly and negatively affect students' satisfaction and performance. Technostress creators indirectly affect students' performance via their satisfaction. We believe that there are several possibilities for the above results. First, many Indian students are not familiar with technology-enabled teaching, they are used to the traditional teaching-learning process, and so they find it hard to disconnect from traditional methods. Second, students do not have sufficient training or knowledge in using technology-enabled teaching tools, which requires additional time and effort from their side. Third, a significant number of students live in rural parts of India, which does not have sufficient infrastructures to support technology-enabled teaching tools; however, many technology-enabled teaching tools can be accessed through mobile devices. Still, other infrastructure facilities like network speed and hardware devices can be an issue. Since technology-enabled teaching is an entirely new concept, students need to get out of their comfort zones to adopt new teaching and learning methods. These findings are seen to hold true in various industry settings (Ragu-Nathan et al., 2008; Tarafdar et al., 2010, 2011, 2015, 2019).

Theoretical Implications

Our findings make some theoretical contributions, and practical implications. The first contribution is in terms of validating and modelling the mediating role of technostress creators on the relationship between technostress inhibitors and students' outcomes. The components of technostress inhibitors, technostress creators, and students' outcomes are validated as separate measurement models, which can be used as a reliable tool to measure the technostress inhibitors, technostress creators, and students' outcomes. Many studies have attempted to find the link between technostress inhibitors, technostress creators, and employee outcome in terms of employee engagement, commitment, satisfaction, and performance (Fuglseth & Sørebø, 2014; Ragu-Nathan et al., 2008; Tarafdar et al., 2010, 2011, 2015, 2019). However, no academic work has tried to find a similar relationship in student populations, particularly during a worldwide crisis.

As the second contribution, the structural model concludes that students' opinions about the technostress supportive mechanisms negatively stimulate their emotional responses leading to stress, while this technostress negatively predicts student satisfaction and end-user performance. In general, this study supports the S-O-R model, and that it can be used in a technology-enabled educational setting. Thus, this study makes a significant theoretical continuation to work in higher education settings. The structural model holistically presented the problem by explaining its possible antecedents and outcomes, which will give better insight to future researchers who would like to research on technostress in a higher educational setting.

As the third contribution, this study explains the negative impact of technostress inhibitors on technostress creators. Institutions can implement the mitigating strategies explained in this study to reduce stress created by the technology-enabled teaching methods.

Finally, interestingly, our result demonstrates the controversial non-significant direct relationship between technostress inhibitors and students' outcomes, such as end-user satisfaction and performance. Technostress inhibitors are generally viewed as a supportive mechanism used to reduce students' technostress., and students do not perceive them as related to their satisfaction and performance. However, in the organisational setting, these supportive measures help to achieve job-related outcomes, which differ from students' outcomes. Our findings show that, in the higher educational setting, students' performance and satisfaction can be improved by reducing the level of stress created by technology by the effective implementation of supportive measures. Prior studies have not addressed this mediating effect of technostress.

Practical Implications

The current study has a few practical implications. It empirically demonstrates the dimensions of technostress inhibitors, technostress creators, and students' outcomes. When students experience a high level of technostress caused by the implementation of technology-enabled teaching, they

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may want to distance themselves from technology-enabled teaching tools (Qi, 2019; Wang et al., 2020). Implementing technology-enabled teaching methods immediately without creating the right environment will lead to technostress (Chauvin et al., 2014; Fugate et al., 2011). However, the current pandemic has mandated immediate switching to technology-enabled teaching without the necessary interval to first prepare the ground for its successful acceptance and adoption. This study recommends that higher education institutes and policymakers understand students' desire for a particular level of technology usage in education, before implementing any technology-enabled teaching methods.

Furthermore, this result explains the importance of the IT help desk, ICT literacy, and motivation towards end-user involvement. Institutions of higher education can use the IT helpdesk to seek support for the issues they face while accessing technology-enabled teaching tools. Based on the students' and educators' level of ICT literacy, different levels of training sessions need to be organised with proper feedback mechanisms. Institutions also need to create a culture that motivates educators and students to try new technologies in the educational field. Implementing reward strategies will also encourage effective adoption of technology.

FUTURE SCOPE OF THE STUDY AND CONCLUSION

This study has a few limitations. We recommend using different student outcomes, such as' academic involvement and motivation as factors affecting technology adoption. This study measures end-user performance in terms of students' productivity and ability to innovate, using self-reported online survey data to prove the relationship between technostress inhibitors, technostress creators, and students' outcomes. Data was collected during the COVID-19 outbreak; however, data collected during normal time could add to the results. A comparative study between developed and developing countries will yield better results. This study is based on the S-O-R model; however, future studies need to consider other theoretical models to develop more comprehensive models since the field of ICT is continuously evolving. The impact of students' demographic profiles was not addressed in this study; it is recommended to consider different demographic profiles to assess the level of technostress.

To conclude, technology has revolutionised the way teachers teach students. The current COVID-19 outbreak necessitated educational institutions to use technology-enabled teaching methods to deliver the courses. It is essential to understand the relationship between technostress creators, technostress inhibitors, and students' outcomes. In this study, we show that technostress inhibitors can negatively influence technostress creators and positively indirectly influence students' satisfaction and their performance.

Conflicts of Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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APPENDIX A

Table 9. Factors and statements

Factors	Statements	Outer loading
Technostress inhibitors	**Literacy facilitation (M=2.29; SD=0.88)**	
	Our college encourages knowledge sharing to help deal with new technology	0.660
	Our college emphasizes teamwork in dealing with new technology-related problems	0.676
	Our college provides end-user training before the introduction of new technology	0.790
	Our college fosters a good relationship between IT department and end users	0.824
	Our college provides clear documentation to end users on using new technologies	0.839
	Technical support provision (M=2.47; SD=0.91)	
	Our end-user help desk does a good job of answering questions about technology	0.836
	Our end-user help desk is well staffed by knowledgeable individuals	0.799
	Our end-user help desk is easily accessible	0.699
	Our end-user help desk is responsive to end-user requests	0.698
	Involvement facilitation (M=2.57; SD=0.93)	
	Our end users are encouraged to try out new technologies	0.913
	Our end users are rewarded for using new technologies	0.744
	Our end users are consulted before introduction of new technology	0.767
	Our end users are involved in technology change and/or implementation	0.634
Technostress creators	**Techno-overload (M=2.31; SD=0.87)**	
	I am forced by this technology to work much faster	0.803
	I am forced by this technology to do more work than I can handle	0.754
	I am forced by this technology to work with very tight time schedules	0.733
	I am forced to change my work habits to adapt to new technologies	0.853
	I have a higher workload because of increased technology complexity	0.837
	Techno-invasion (M=2.18; SD=0.85)	
	I spend less time with my family due to this technology	0.734
	I have to be in touch with my work even during my vacation due to this technology	0.876
	I have to sacrifice my vacation and weekend time to keep current on new technologies	0.777
	I feel my personal life is being invaded by this technology	0.771
	Techno-complexity (M=3.02; SD=1.01)	
	I do not know enough about this technology to handle my task satisfactorily	0.850
	I need a long time to understand and use new technologies	0.873
	I do not find enough time to study and upgrade my technology skills	0.678
	I find other students in this college know more about computer technology than I do	0.634
	I often find it too complex for me to understand and use new technologies	0.784
	Techno-insecurity (M=2.39; SD=0.80)	
	I feel a constant threat due to new technologies	0.749
	I have to constantly update my skills to avoid being replaced	0.786
	I am threatened by co-workers with newer technology skills	0.744
	I do not share my knowledge with my co-workers for fear of being replaced	0.835
	I feel there is less sharing of knowledge among co-workers for fearing of being replaced	0.763
	Techno-uncertainty (M=2.61; SD=0.90)	
	There are always new developments in the technologies we use in our college	0.741
	There are constant changes in computer software in our college	0.746
	There are constant changes in computer hardware in our college	0.838
	There are frequent upgrades in computer networks in our college	0.827

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Table 9. Continued

Factors	Statements	Outer loading
End-user satisfaction	**Content (M=2.02; SD=0.73)**	
	The system provides the precise information I need	0.789
	The information content meets with my needs	0.807
	The system provides reports that seem to be just about exactly what I need	0.774
	The system provides me with sufficient information	0.739
	Accuracy (M=2.11; SD=0.68)	
	The system is accurate	0.770
	I am satisfied with the accuracy of the system	0.828
	Output (M=2.23; SD=0.92)	
	I think the output is presented in a useful format	0.799
	The information provided by the system is clear	0.775
	Ease of use (M=2.08; SD=0.67)	
	The system is user friendly	0.822
	The system is easy to use	0.827
	Timeliness (M=2.04; SD=0.63)	
	I get the information I need in time	0.840
	The system provides up-to-date information	0.736
End-user performance	**ICT-enabled productivity (M=1.54; SD=0.73)**	
	This technology helps to improve the quality of my work	0.727
	This technology helps to improve my productivity	0.789
	This technology helps me to accomplish more work than would otherwise be possible	0.755
	This technology helps me to perform my task better	0.832
	ICT-enabled innovation (M=1.86; SD=0.67)	
	This technology helps me to identify innovative ways of doing my task	0.838
	This technology helps me to come up with new ideas relating to my task	0.816
	This technology helps me to try out innovative ideas	0.811

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Blended Learning in Teaching Technical Teacher Candidates With Various Types of Learning Styles

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ABSTRACT

To compare the different impacts of the balance of face-to-face and online learning in blended learning, along with learning styles, an experiment was done using a 3x4 design consisting of three blended learning composition groups of 25% face-to-face and 75% online, an equal balance of face-to-face and online, and 75% face-to-face and 25% online. There were four learning style type groups of Diverger, Assimilator, Converger, and Accommodator. The population was student teachers in electrical engineering. Students of each style were randomly allocated to the three blended learning groups. It was found that both the blend of online and face-to-face learning and the types of learning styles affect competence outcomes significantly in some combinations.

KEYWORDS

Accommodator, Assimilator, Competence, Converger, Diverger, Electrical Engineering, Face to Face, Learning Style, Online, Technical Teachers

INTRODUCTION

The combination of online and face-to-face combination - blended learning – has been claimed to have many advantages (Nazar, Omer, Nazar, & Husband, 2019; Basogain, Olabe, Ifinedo, Pyke, & Anwar, 2018) because the two modes can be mutually reinforcing, with each complementing the other (Dendir, 2018; Yang, Yu, & Chen, 2019). Moreover, the model is able to develop quality teaching (Kennedy, 2016; Pekkarinen & Hirsto, 2017; Cutri & Whiting, 2018) through the pedagogical properties of student-centered and collaborative learning (Wong et al., 2018; McIlveen, 2018; Parsons, Ankrum, & Morewood, 2016). Blended learning can help to apply inquiry-based learning, problem-based learning, and project-based learning related to authentic professional practices, phenomena, problems, and situations.

Digital and mobile communication can make content interactive and lessons adapted to the cultural preferences of students. Students can adapt (Ruhalahti, Korhonen, & Rasi, 2017) by transferring knowledge and work while learning takes place (Hortovanyi & Ferincz, 2015) through Information and Communication Technology (ICT) (Ruhalahti, Aarnio, & Ruokamo, 2018; Khusainova & Lukoyanova, 2018). The learning becomes more dynamic, interactive, and motivating (Cuesta Medina, 2017; Boelens, Voet, & De Wever, 2018; Hughes, 2007; Chmiel, Shaha, & Schneider, 2017;

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Broadbent, 2017), and can foster independent of learning and rich understanding (Nickels & Gartner, 2018). The blended model can also give students to cognitive presence and social experience through synchronous or asynchronous discussions with colleagues and facilitators (Donnelly, 2006; Ndlovu & Mostert; 2017).

Nevertheless, there is a finding that learning outcomes through blended learning do not differ significantly from traditional learning (Botts, Carter, & Crockett, 2018). Moreover, some statements contradict each other that claim learning styles do not affect learning achievement (Brunton, 2014; Kirschner, 2017), but others do not fully agree (Martinez & Tuesday, 2019; Toyama & Yamazaki, 2019; Huang, 2018). However, we think it is necessary to explore that information to consider implementing blended learning while considering the impact of different learning styles on blended learning implementation. Other factors might play a role in influencing the learning process through blended learning and should also be investigated. Moreover, more information about the form of integration most suitable for each type of learning style is needed - in this study teacher candidates in electrical education are the study cohort. Generally, these students come from various situations and conditions, families, home areas, and types of education and economic levels. Hence, we are interested in investigating the use of blended learning to teach them to obtain maximum learning outcomes. The form of online and face-to-face integration is a focus attention of this study because the diversity of integration modes may have different impacts on each group of study participants. The study takes account of claims that the success of blended learning is not the simple integration of teaching in the classroom with digital media but changing patterns and practices of learning (George-Walker & Keeffe, 2010; López-Pérez, Pérez-López, & Rodríguez-Ariza, 2011). Likewise, the diversity of student characteristics may also contribute to differences in the results obtained and differences in the learning material's characteristics (Kim McShane, 2004; Alebaikan & Troudi, 2010). We focus on learning styles because they play an essential role in the learning process (Chang-Tik, 2017; Green & Sammons, 2014). The impact of different learning styles and forms of integration on outcomes are essential to investigate because different individuals in different conditions gave different results (Premlatha, Dharani, & Geetha, 2014; Yang & Quadir, 2018; Shen & Palmeri; 2016; He, 2005).

Building competencies in electrical teacher candidates is essential because they have become necessary provisions to involve them in further teaching assignments (Zhu, Wang, Cai, Engels, 2013; Kömür, 2010; Baran, Correia, & Thompson, 2011) and master the theory and practice of electricity (Keltikangas & Martinsuo, 2009) for planning, implementation, and evaluation of learning (Goodwin & Kosnik, 2013; Kantonidou & Chatzarakis, 2005). This paper describes how teacher candidates' competency is built through appropriate forms of blended learning.

LITERATURE REVIEW

The incorporation of online and face-to-face learning can provide students with excellent opportunities to interact with fellow students and lecturers online (Shorey et al., 2018; Ra, Chin, & Lim, 2016). The convergence between technology-based environments and traditional arrangements is recognized as an effective model and has been implemented in a virtual laboratory in flipped classrooms (Chapman & Hassein, 2018; Gilboy, Heinerichs, & Pazzaglia, 2015). In this context, students can get learning assistance in the form of guidance or demonstrations/simulations, both directly and indirectly (Vitale & Linn, 2018; De Jong, Linn, & Zacharia, 2013). Therefore, it is possible to support independent learning and collaboration (Stoltzfus, 2016; Lim, Yan & Xiong, 2015) because more communication channels are supported (Kimmelmann & Lang, 2018).

Students' speed in processing information varies (Alexakis & Andert, 2015; Kurtz et al., 2013), so choosing cooperative learning for each group is essential. A variety of online and face-to-face blends is possible to fulfill the learning needs. The provision of learning assistance is appropriate to each group based on shared individual learning characteristics (Haruehansawasin & Kiattikomol, 2017; Shu & Gu, 2018). Online learning can provide opportunities for prospective teacher students

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who are fast learners and have high learning creativity. In contrast, face-to-face learning can assist the slower learners and provide encouraging results (Lockhart, McKee, & Donnelly, 2017; Kent, Laslo, & Rafaeli, 2016; Ansari Ricci, Persiani, Williams, & Ribas, 2019; Wetzel, De Arment, & Reed, 2015). The study time for students independently and with lecturer involvement in the classroom can be arranged as needed. In shaping the professional prospective teachers' competence, teaching material is delivered through blended learning, which combines online and face-to-face instruction (Jean in Inoue, 2010). The prospective electrical engineering teacher needs to understand the concepts of electricity and have skills in selecting and using the appropriate tools on the task, efficient use of time in carrying out tasks, work safety, and precision of the work results.

In this study, we make variations in three types of combinations of online and face-to-face instruction, adjusted for the learning material. The three types are 25% online and 75% face-to-face instruction, 50% online and 50% face-to-face instruction, and 75% online and 25% face-to-face instruction. IThese relate to the individual's speed difference in digesting information. Diversity of integration is intended to be able to accommodate the diversity of differences that students have. Online learning provides broad opportunities for students to learn independently, especially those who have high learning creativity, and face-to-face can be effective in facilitating and motivating slower students (Wilke, King, Ashmore, & Stanley, 2016; Zwart, Van Luit, Noroozi, & Goei, 2017). Thus, their different competencies can be developed maximally because it is possible for these differences to be accommodated (Manwaring, Larsen, Graham, Henrie, & Halverson, 2017; Kuo, Belland, Schroder, & Walker, 2014), as monitored through measurement and assessment (Moss & Brookhart, 2009).

Learning styles are a process where knowledge is formed through the transformation of experience. Based on Kolbs' learning experience theory, it starts from the Concrete Experience stage (CE), Reflective Observation (RO), Active Experiments (AE), and Abstract Conceptualization (AC). Teaching materials refer to delivering blended learning practices so that students with various learning styles are served. The bipolar continuum (CE and AC) is orthogonal to the bipolar continuum of both RO and AE, which are contiguous modes of preference from the experiential learning cycle. They lead to four basic learning style combinations, which are known as Diverger (between CE and RO), Assimilator (between RO and AC), Converger (between AC and AE), and Accommodator (between AE and CE). Someone may have one of these preferred styles but can learn in and use the other modes. Divergers have strong imaginative abilities and are good at using various views to see things creatively and work with others. Those who prefer inductive thinking and abstract ideas and make theoretical models are Assimilators. Convergers have a strong practical orientation; they are generally deductive and tend not to be emotional, while Accommodators love to do things and solve problems intuitively and take risks (Manolis, Burns, Assudani, & Chinta, 2013).

The learning design uses various approaches to accommodate diverse student learning modes (Pizzi, 2014). Some people say that non-traditional students generally prefer to start from the AC-AE quadrant, whereas traditional students prefer the RO-AC quadrant (Svinicki & Dixon, 1987; Vince, 1998; Wynd & Bozman, 1996). These variations illustrate that using a variety of face-to-face instruction and online instruction also needs to be adapted to different learning styles. For example, more online learning may be more suitable for convergent types. For this reason, it is necessary to choose the right form of learning to give maximum results - that is, high student performance in the course (Chimmalgi, 2018; Vizeshfar & Torabizadeh, 2018; Vanslambrouck, Zhu, Lombaerts, Philipsen, & Tondeur, 2018).

METHOD

Tools and Materials

We built learning webs (URL: http://jpte-ft-unimed.edu20.org) as a platform for this study. Validating completeness of navigation, information loading space, the display aspects, and ease of use of the site was done by involving IT and learning technology support. The learning material prepared is

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referred to as the curriculum, and involved three experienced lecturers in assessing the conformity aspects of the material content. Mastery theory tests and skills assessment sheets as competency test instruments were developed, referring to learning objectives. Theory mastery was measured using 25 questions in the form of essays. Dexterity using tools on the task, choosing the right tool, the precision level of the work results, time-efficient use in carrying out tasks, and work safety were all included in the practical skills assessment sheet. The theory mastery scores were determined by giving a score of two if correct and zero if incorrect for all items, so the maximum total score was 50. The range score of Skill Assessment is 0 – 10 for each aspect. Therefore, the maximum total score also 50. Then, the competency score is a combination of the two scores. We tested the instrument's validity first by conducting a test that involved 35 participants and three raters (the lecturers in electrical engineering). No significant differences were found between them (F=.145, Sig,=.865) and we concluded it was feasible to use. We then uses Kolb's Learning Style Inventory 4.0 for tracking participants' learning style types,

Participants and Design

Eighty-four people were selected to become participants in the experiment using the Kolb Learning Style Inventory 4.0. They consisted of 21 people for each different learning style, namely Diverger (Di), Assimilator (As), Converger (Co), and Accommodator (Ac). A total of seven participants were randomly drawn from each learning style group to be placed in a study group with the three different blends of face to face and online learning (25%-75%, 50%-50% and 75%-25%) so that the experimental group became a 3 x 4 design.

Procedure

Pretests were given in advance to all experimental groups to find out their prior knowledge base. Based on the analysis, their initial competencies did not differ significantly (F=2.149, Sig.= 0.76 > .05). The learning activities were then carried out. One lecturer in each group taught activities up to 12 times in one semester. When learning online, the participants had different access codes for each learning group. The groups were BL1 (25 – 75 blended, four times online and eight times face to face); BL2 (50 – 50 blended, six times online and six times face to face); and BL3 (75 – 25 blended, eight times online and four times face to face), Each blend covered the same time and material. As soon as the learning ended, competency tests were carried out and each group member's competency data were recorded.

Data Analysis

The descriptive statistics used to describe competency data were the Kolmogorov Smirnov Test and the Levene Test use to test normality and homogeneity of data. The variety of learning styles, the different forms of learning, and the interaction of learning with learning styles and their effect on competencies was tested through two-way ANOVA at the significance level of .05. Moreover, differences in the average of competencies between learning groups and the type of learning style group was measured through post hoc testing. Plotting the competencies of each group's type of learning style in each form of learning suggests the most suitable combination. All data analyses were carried out through IBM's SPSS 25 program.

RESULTS

Competence Description

The participants' competency description of each learning group (n = 28) shows that the 25-75 Blended Group had the highest score on average (Mean = 73.68; Sd = 7.977), followed by 50-50 Blended (Mean = 75.61; 6.437), and the 75-25 Blended Group (Mean = 71.04; Sd = 5.818). Based

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on the learning style group scores, the Assimilator type group members had higher scores (Mean = 76.33; Sd = 7.492) than the members of the other groups: Diverger (Mean = 74.76; Sd = 9.006), Accommodator (Mean = 72.19; Sd = 2.294), and Converger (Mean = 70.48; Sd = 7.336).

Regarding competence in learning by learning style type (n = 7), the Divergers' competency scores showed the highest average score in the 25-75 blended learning group (Mean = 83.29; Sd = 2.690), followed by the 50-50 blended group (Mean = 77.86; Sd = 2.911) and the 75-25 blended group (Mean = 63.14). The average score of the Assimilator group was highest in the 50-50 blended group (Mean = 84.00; Sd = 2.309), followed by the 25-75 blended group (Mean = 76.86; Sd = 4.670) and the 75-25 blended group (Mean = 68.14; Sd = 3. 579). The Converger learning style types had the highest competency score in the 75-25 blended group (Mean = 79.29; Sd = 3.773), followed by the 50-50 blended group (Mean = 68.43; Sd = 2.820) and the 25-75 blended group (Mean = 63.21; Sd = 3.147). Meanwhile, the Accommodator type had the highest competency score in the 75-25 blended group (Mean = 72.57; Sd = 2.070), followed by the 50-50 blended group (Mean = 72.14; Sd = 1.574) and the 25-75 blended group (Mean = 70.86; Sd = 1.676).

Regarding the participants' competency score obtained in the learning group by learning style type (n =7), the Diverger type had the highest scores in the 25-75 blended learning group (Mean = 83.29; Sd = 2.690) followed by the Assimilator (Mean = 76.86; Sd = 4.670), Accommodator (Mean = 70.86; Sd = 1.676), and Converger types (Mean = 63.71; Sd = 3.147). Participants' competency scores in the 50-50 blended learning group, sorted from the highest to lowest scores, are Assimilator (Mean = 84.00; Sd = 2.309), Diverger (Mean = 77.86; Sd = 2.911), Accommodator (Mean = 72.14; Sd = 1.574), and Converger types (68.43; Sd = 2.820). In the 75-25 blended group, the converger type scored the highest (Mean = 79.29; Sd = 3.773), followed by the Accommodator (Mean = 73.57; Sd = 2.820) , Assimilator (Mean = 68.14; Sd = 3.579), and Diverger types (Mean = 63.14; Sd = 2.193).

Data normality is fulfilled (N = 84, Mean = 71.79, Std. Dev. = 6.174, Test statistic = .089), and (Asymp.Sig (2-tailed) = .113 > .05). Also, data homogeneity based on the Mean, Median, Median, and adjusted df, and the Trimmed Mean is Sig.> .05 so analysis continued.

Blended Learning's and Learning Styles' Influence

The various forms of blended learning, learning styles, and their interactions influence the competence of the prospective teacher significantly (Intercept, F = 51381.495, DF = 11, Sig. = .000 < .05). Then Blended learning-BL varieties have a significant influence on competence (F = 16,725, df = 2, Sig.=.00 <.05) as well as the LS-style Learning variety (F = 16,246, df = 3, Sig. = . 00 < .05). Also, their interaction (BL * LS) is significant (F = 57,219, Sig.= 00 < .05). There are significant competence differences among blended learning groups (Table 1).

Several pairs of groups are not significantly different among groups of learning style types (Diverger-Assimilator, Mean Difference = .157; Sd = .916; Sig.=.324 > .05; Converger-Accommodator, Mean Difference = 1.71, Sd = .916, Sig. =. 250 > .05) while other pairs are significant (Table 2).

Figure 1 depicts the competency of each group based on blended forms and learning styles combination. It appears as though people with the diverger learning-style type are better in 25-75 blended learning than assimilators, converters, and accommodators, who performed better, respectively, in 50-50 blended, 75-25 blended, and 75-25 blended.

DISCUSSION

The diversity of online and face-to-face learning combinations gave different impacts on teacher candidates' competencies, in line with the findings of Kim McShane (2004). The composition of blends is fundamental for considering blended learning implementation, as seen in the findings of López-Pérez, Pérez-López, & Rodríguez-Ariza (2011). Using blended learning is not only based on the integration of classroom teaching with digital media but also needs to consider other factors to

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Table 1. Competence comparison among blended learning groups

(I) Blended	(J) Blended	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
25-75 Blended	50-50 Blended	-1.93*	.794	.046	-3.83	-.03
	75-25 Blended	2.64*	.794	.004	.74	4.54
50-50 Blended	25-75 Blended	1.93*	.794	.046	.03	3.83
	75-25 Blended	4.57*	.794	.000	2.67	6.47
75-25 Blended	25-75 Blended	-2.64*	.794	.004	-4.54	-.74
	50-50 Blended	-4.57*	.794	.000	-6.47	-2.67
*. The mean difference is significant at the .05 level.						

make blended learning superior in practice to create diverse learning patterns and learning materials (George-Walker & Keeffe, 2010).

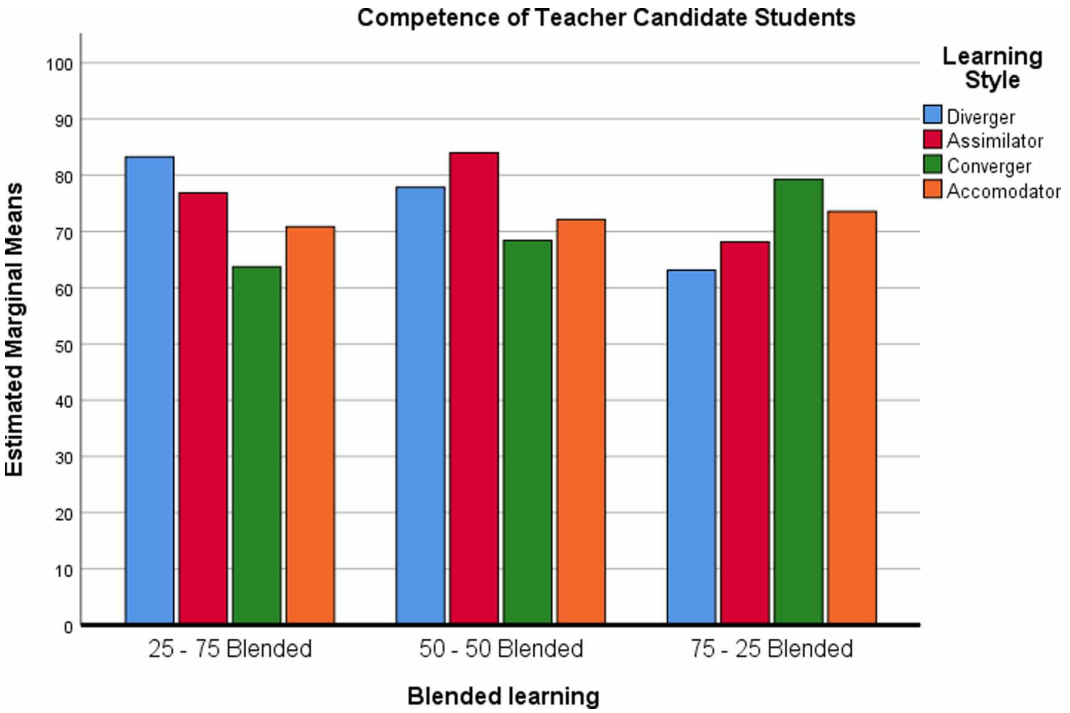
In this experimental condition, several groups showed that competence based on learning styles did not significantly differ, especially with similar types, namely, the diverger type group against the assimilator type group and the converger type group against the accommodator type group. This further reinforces finding of Huang (2018) that not all learning styles make a significant difference in academic achievement. This is possibly because of the similarity of learning occurring between the process of concrete experience and reflective observation (diverger type) and the process of reflective observation and active experimentation (assimilator type). The same thing happens between the active experimentation process and abstract conceptualization (converger type). This borders on the process of abstract conceptualization and reflective observation (accommodator type). The diverger group

Table 2. Competence comparisons among learning style groups

(I) LS	(J) LS	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Diverger	Assimilator	-1.57	.916	.324	-3.98	.84
	Converger	4.29*	.916	.000	1.88	6.70
	Accomodator	2.57*	.916	.032	.16	4.98
Assimilator	Diverger	1.57	.916	.324	-.84	3.98
	Converger	5.86*	.916	.000	3.45	8.27
	Accomodator	4.14*	.916	.000	1.73	6.55
Converger	Diverger	-4.29*	.916	.000	-6.70	-1.88
	Assimilator	-5.86*	.916	.000	-8.27	-3.45
	Accomodator	-1.71	.916	.250	-4.12	.70
Accomodator	Diverger	-2.57*	.916	.032	-4.98	-.16
	Assimilator	-4.14*	.916	.000	-6.55	-1.73
	Converger	1.71	.916	.250	-.70	4.12
The error term is Mean Square(Error)=8,512. An asterisk (*) signifies that the mean difference is significant at the .05 level.						

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Figure 1. Teacher candidates' competence based on learning style in blended learning



competences were very different from the converger group, as were the assimilator and accommodator group's competences. This is in line with previous research - Toyama & Yamazaki (2019) and Martinez & Tuesca (2019) state that different learning styles provide significantly different learning outcomes.

Based on the results from the learning groups, the diverger learning style group showed higher competence in the 25 percent online and 75 percent of face-to-face learning combination, while the converger group befitted from the 75 percent online and 25 percent face to face combination. This suggest that students with the diverger learning style benefit from learning that is more directed towards traditional learners, while the converger type group is non-traditional as stated by Svinicki & Dixon (1987), Vince (1998), and Wynd & Bozman (1996). The assimilators were better at learning with a composition of 50 percent online and 50 percent face-to-face learning, and the accommodator type did better with the 75 percent online and 25 percent face-to-face learning. This shows that different individuals in different conditions tend to produce different results in line with Yang, & Quadir (2018), Shen & Palmeri (2016), She (2005), and Premlatha, Dharani, & Geetha (2014).

CONCLUSION

This research considered the balance on online and face-to-face learning in blended learning to obtain maximum results in student teachers. The results suggest that this balance of online and face to face learning has an impact, but so do different types of learning styles. The interaction between the learning styles of prospective teachers and blended learning was also found to be significant. Teacher candidates' resulting competencies based on their type of learning style were different in each online and face-to-face blended group. So, for teaching them, it is necessary to pay attention to their type of learning style first to choose the right approach and obtain maximum learning outcomes.

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IMPLICATIONS

Learning styles are found to play an essential role in the process of forming teacher competency. Therefore, forming learning groups based on students' learning styles should be considered so that the results obtained can eb maximized. The learning styles of students are therefore identified first, for that purpose.

LIMITATIONS

This study only focused on three formats of online learning and face-to-face learning in the context of blended learning. These compositions were applied to prospective teacher students (in the field of electrical engineering) with various types of learning styles. Based on these limited conditions, subsequent researchers have an opportunity to review other factors that play a role in the teaching and learning process through blended learning, perhaps investigating the role of other learning characteristics, including cognitive style and cognitive control.

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Conflicts of Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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Gamified E-Reading Experiences and Their Impact on Reading Comprehension and Attitude in EFL Classes

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ABSTRACT

The use of computers and electronic devices for recreational reading and for reading in educational settings has gone up significantly in recent years. Whereas the digital revolution is rapidly changing the world, it is also changing education. This study examined the perceptions of secondary school EFL learners in Turkey of their e-reading experiences based on their gamified electronic reading practices in school and their influence on reading comprehension performance in an EFL class. The findings revealed that the implementation of e-book reading resulted in higher comprehension levels and more positive reading attitudes. Participant students showed a preference for printed books rather than electronic books for leisure due to the sense of ownership that the printed text storybooks offered. However, the results indicate that EFL learners' use of screen reading has the potential to improve students' attitudes towards reading in educational settings.

KEYWORDS

EFL, Electronic Reading, Gamification, Gamified E-Reading, Reading Attitude

INTRODUCTION

A technologically literate population is a hallmark of any nation in pursuit of sustainable and successful growth, and reading plays an essential role in this context as it is vital to anyone who wishes to learn a skill; it allows individuals to remain on top of current affairs and this steers life trajectory and empowers the wider community to participate in civic engagement (Mullis et al., 2011). Scholars in the field of education hold this vision. They never stop studying and discovering ways to improve one's reading ability in their effort to extend the area of their expertise (Mckee, 2012; Mokhtari & Richard, 2002; Mullis, et al., 2011).

Gamification has become a trending topic in education because of its impact on student learning (Göksün & Gürsoy, 2019). Gamification is the technique of adding game elements to non-game contexts (Deterding et al., 2011). The goal of gamification is not to create a new game-like environment, but to move the elements of the game into the real world to catch similar senses without leaving the truth (Arkün-Kocadere & Samur, 2016, pp. 397–414). Gamification in education is a way of playing imaginative games in the classroom without endangering a curriculum's scientific nature (Nolan & McBride, 2014). Gamification helps the individuals in an educational setting to develop critical thinking and multi-tasking by educating effective digital natives of the 21st century (Kapp, 2012; Prensky, 2014). Gamification also turns the learning experience into a more enjoyable one and increases the students' motivation to learn and study (Muntean, 2011). Moreover, gamification

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offers student learning data by allowing more accessible, reliable, and timely information for students, parents, administrators, and policymakers (Darling-Hammond, 2010).

Gamified e-books offer incentives to young readers to explore the world of literature. The power of gamification to incentivize engagement has not been recognized by most educators. Khan Academy, a free to use learning resource containing lectures on subjects from mathematics to music, utilizes the badge system to illustrate subject mastery. It also encourages the "students" of Khan Academy to return daily, by awarding achievements for logging in to the website over a certain period. Public schools and universities are also following suit. Teachers are incorporating badge systems into the classroom to reward performance achievements. McGraw Hill Publishing's "Connect" platform has online activities to accompany textbooks; these activities often display a score and leaderboard, so students can compare the points they've earned to their classmates' scores or their own personal best. The use of achievements has been shown to improve student performance by allowing them to reach tangible goals before the end of the course (Ames, 1992). The literature of gamification in EFL classes is limited and there is not enough research about the influences of gamification in education to increase learning outcomes, particularly in reading skill. Therefore, the current study seeks to identify the potential effects of gamification on Turkish EFL sixth-grade students' reading comprehension and digital attitudes.

GAMIFICATION IN EDUCATION

Secondary school learners' reading abilities are at risk as they have experienced repeated struggles and tend to adopt a passive role during instruction all around the world (Zimmerman & Schunk, 2006). In contrast, when the literature is analyzed, gamification has been used for numerous motives in educational backgrounds with positive learning results (Millis et al., 2017; Noroozi, McAlister, et al., 2016).

Learners frequently conveyed that having gamified EFL settings is pleasing, enjoyable, interactive, and stimulating (Baldauf, Brandner, & Wimmer, 2017; Guaqueta & Castro-Garces, 2018; Homer, Hew, & Tan, 2018; Medina & Hurtado, 2017; Sun & Hsieh, 2018), as it gives the students a chance to get involved in the learning process (Guaqueta & Castro-Garces, 2018). According to Morford et al. (2014), elements of game playing (e.g., a player's direct impact on the game's results, clear end goals, rules for play, and development of strategies to complete the tasks) can serve to motivate students in three different ways. First, these gaming elements can be socially motivating by offering opportunities for collaboration with team members or competition with other players. Second, academic games are emotionally motivating when students earn rewards and receive feedback because they become immersed in the game's tasks. Finally, to instantiate these properties, the game must be sufficiently challenging to engage students without frustrating them.

Gamification has also been used for assessment. The primary purpose of adding game elements is to better engage the test taker (Armstrong et al, 2016), but by changing students' attitudes and behaviors, gamification could affect assessment outcomes in ways not well understood by the test developers (Landers, 2014). This may be why gamified assessments tend to be simpler and use basic game elements such as 2-D graphics and sound effects; whereas training games are more likely to resemble high-tech, commercial video games such as 3-D graphics and role-playing (Armstrong et al., 2016).

Researchers have advised test developers to add game elements iteratively, starting with a few and gradually increasing or refining them (Landers, 2014). This allows investigating how the elements are affecting test takers—including those with disabilities—and the test's validity. Previous studies have focused on quantitative factors that mediate or moderate students' performance on computerized assessments such as grade or age, gender, special education, attitudinal ratings, and categorical designations for computer features.

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GAMIFICATION IN FOREIGN LANGUAGE CLASSES

When it is compared with traditional electronic books, learning with gamified electronic books has become a gradually more striking alternative, which can attract the interest of learners. Gamification is a term that has acquired ample attention (Deterding et al. 2011; Kapp, 2012; Lee & Hammer, 2011). Game mechanics can be defined as the rules that control the player's moves and the game's response to the players. According to Kapp (2012), points, levels, challenges, feedback and leaderboards are the most common game mechanics.

Points / levels: Instead of giving concrete rewards, when players perform the challenges in the game successfully, they receive points, they think they are rewarded. Gamified electronic books generally have interactive point systems. The reward systems which are integrated in digital games are one of the significant reasons for player interaction (Reeves & Read 2009). Zagal et al. (2005) categorize points and levels as goal metrics, they keep track of the game and show progress on player performance.

Challenges (with badges, achievements, and trophies): The introduction of challenges in an activity makes students (users) feel like they are working toward a goal. Csikszentmihalyi (1990) claims that challenges should be organized according to the user's ability, they should not be too hard or too easy. If the challenge is lower than the ability of the player, it results in boredom and detachment, whereas when it is much higher than the player's ability, it creates anxiety, stress and ultimately resignation (Csikszentmihalyi, 1975). If the challenge is high but within the capacity of the player, the result is a state of flow - extreme absorption, enjoyment and feelings of achievement (Csikszentmihalyi, 1990).

Feedback: Feedback plays an essential role in improving and sustaining the cognitive engagement of learners (Gresalfi & Barnes, 2016). Receiving feedback reduces the cognitive load of the player and decreases the uncertainty condition inside of the game (Chang et al. 2017).

Leader boards or "high-score tables": Leaderboards not only motivate certain players to continue to illustrate honor over achievement (Werbach & Hunter, 2012; Yee, 2006), these game elements also cause competition, and fear (Mekler et al., 2013).

In educational settings, scoring, stars, and points have always been used by educators in classes. As a results, points, levels, badges, achievements, feedback, and leaderboards are the most commonly employed game elements in game and non-game contexts.

The use of gamified reading in EFL classes might increase the inner motivation of the learner (Liman Kaban & Karadeniz, 2021). Zhu et al. (2018) also examined the consequence of using gamification on reading literacy. There were twenty-nine students from the elementary stage. The results revealed that gamification influenced learners' reading comprehension positively. On the other hand, Chen et al., (2019) examined the influences of using gamification on fifty-five fifth grade students' reading performance. According to the study, there were no statistically significant differences between groups. Yavuz, Ozdemir, and Celik (2020) claimed that the students who had gamified treatment showed statistically lower anxiety levels than the students using the traditional methods. According to Phuong (2020), students' attitudes towards a gamified learning environment in an EFL class was positive.

"Reading Battle" a gamified activity to strengthen student's reading comprehension, supported active reading practices in students who used the gamified reading, improved their academic performance, enhanced their reading competence, and boosted their reading motivation (Li, Mok, Cheng & Chu, 2018). Liman Kaban and Karadeniz, (2021) could not find statistically significant differences in the reading comprehension scores of sixth grade EFL students in a state school between groups, but they found students who took gamified treatment increased their motivation. It is possible to conclude from these studies that gamified reading can increase the motivation of reading but may have less influence on reading comprehension.

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READING ATTITUDE IN THE DIGITAL AGE

Reading attitude can be defined as the mood about reading that affects the embracing or avoidance of positive reading habits. Research consistently highlights a worsening of students' reading attitudes as they move through school (McKenna et al., 1995). In fact, many students self-identify as nonreaders or resistant readers (Lenters, 2006). It is possible to state that when students label themselves as a non-reader, they also show a digital attitude against reading. It is necessary to contemplate students' digital attitudes for various reasons. Reading in digital mode means the comprehension of codes illustrated digitally. Different from reading on traditional paper, reading electronic texts helps the reader make choices of entry points into a text and the selection of paths to follow (Rhodes & Robnolt, 2009). Research on the relationships between the increased use of electronic reading and outcomes is in the early stages. The purpose of this paper is to investigate adolescents' reading by investigating four modes of reading: 1) academic paper mode, 2) academic digital mode, 3) recreation paper mode and 4) recreation digital mode.

METHODS

The main purpose of this study is to determine the influences of gamified electronic book reading on reading attitude levels of sixth grade EFL students at a private school in Istanbul, Turkey. The secondary purpose of the study is to analyze secondary school English teachers' opinions about their experience in using a gamified electronic book platform. In this study, both quantitative and qualitative data collection instruments were used. In this quasi-experimental study, data were gathered through a 5 point Likert-type Adolescent Reading Attitudes Survey (ARAS) and teacher / student focus group interviews. Random sampling was not used in this design due to feasibility reasons. The groups that were present were matched for certain features and participants were randomly assigned to the groups. However, this still did not guarantee that these groups were equivalent.

RESEARCH QUESTION

The question "What are the effects of using gamified reading applications on students' reading attitudes in EFL Classes?" forms the basis of the research.

The sub-questions investigated within the framework of the basic question of the study are as follows:

1. What is the impact of gamified reading comprehension activities on students' reading attitude?
2. What are the professional and personal views of teachers on gamification?
    a. What are the views and recommendations of teachers who participated in activities in the gamified linear electronic reading program?
    b. What are the views and recommendations of teachers who participated in activities in the gamified non-linear electronic reading program?
3. What are the perceptions of the students about gamified e-reading?

This study scrutinized these subjects through the data gathered from the tests and scales taken by the participating young learners who had been exposed to gamified online reading lessons and from the perceptions of these students and of their teachers who had implemented the lessons. With the aim of investigating the above questions, this study involved 70 young learners, who were currently sixth graders, and three EFL teachers at a private secondary school in Istanbul, Turkey, and included a qualitative analysis of student and teacher focused interviews and teacher journals to form a solid student and teacher based views on applying gamified reading in and outside the classroom.

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RESEARCH DESIGN

The quantitative part of the study was composed of a reading comprehension placement test that was applied before the implementation to determine the English reading comprehension levels of the participants as a pre-test, a reading comprehension as a post-test, and a Likert-scale Adolescent Reading Attitudes Survey (ARAS) by the student participants both as a pre-test and post-test. As for qualitative data, they were composed of the interviews of the participant teachers conducted by the researcher online and interviews with 15 students chosen randomly among the participants.

As the implementation of gamified online reading lessons in 2 sixth grade classrooms was designed as a specific instance in action to illustrate more general grounds in a real context and its effect was gauged and explored with a control group, the present study can be described as a case study with a quasi-experimental study. To determine the current English proficiency levels of the participant students, a four-skill placement test was conducted in 3 classes simultaneously at the very beginning of the study. To find out whether the treatments lead to increased reading comprehension of the 72 participants, two sets of pre- and post-tests were performed at the beginning and the end of the four-week teaching intervention. Finally, to complement the findings from quantitative tests and questionnaires, interviews with the teachers along with the answers provided by the participant students in the focus group interviews were analyzed qualitatively (Table 1).

SETTING AND PARTICIPANTS

This study was conducted at a private secondary school in Turkey, İstanbul. The EFL program which the sixth graders of the school – the participants of the present study – were exposed to is a four-skill integrated curriculum with 2 lesson hours each week. A private school which had at least three sixth graders class and 24 students in each classroom was found, in which this study was carried out. Two teachers stayed in each class: one Turkish EFL teacher as an observer and the researcher who is also an experienced EFL teacher. The experiment was conducted as an after school online activity and 40 minutes was allocated for each book every week. The ages of the students varied between 11 and 12.

Table 1. Design groups in the study

Groups	Pre- Test	Treatment	Post-Test
Gamified Linear Electronic Reading Program	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS)	Four week reading program was applied with a 40 minute long gamified linear reading session every week. Students read 4 books in total.	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS) Structured Interviews with students / teachers
Gamified Non-linear Electronic Reading Program	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS)	Four week reading program was applied with a 40 minute long gamified non-linear e-reading session every week. Students read 4 books in total.	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS) Structured Interviews with students / teachers
Printed Guided Reading Program	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS)	Four week reading program was applied with a 40 minute long printed guided reading session every week. Students read 4 books in total.	Reading Comprehension Test Adolescent Reading Attitudes Survey (ARAS) Structured Interviews with students / teachers

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PROCEDURES

No names were used throughout the report to ensure confidentiality. Written permission was obtained from the principal. We also received written consent from each of the seventy-two students' parents. The informed consent also gave contact information for the researcher and research advisor.

Part of the data collection instruments includes in-depth analytical procedures that clarify and analyze quantitative and qualitative methods; data collection procedures that illustrate data sources, sampling, method of implementation, and guidance. At the end of the four-week reading program, a post-test was given to see if there was any significant difference in reading attitude of the three groups of EFL students. Students and teachers were interviewed individually with videoconferencing by a trained research team member.

Both the pre-test and the post-test were in the same format of multiple-choice questions from the reading comprehension texts which were created by field experts. For triangulation, a qualitative interview from the respondents had been done to support the results from the statistical findings after the post-test. To answer the research question, an independent and a paired sample t-tests were carried out to study the effects of the employment of the digital reading program on students' reading comprehension of English short prose texts before and after the program. ANCOVA analysis with pretest as a covariate was also employed to check if there is any significant difference in the comprehension pretest and post-test of the EFL readers of the two groups.

MEASURES

Reading Comprehension Tests and Gamified Electronic Reading Platform

Reading comprehension tests were used to assess the students' level of comprehension before and after the study as a pre-test and post-test. Reading comprehension tests were developed to compare reading comprehension of the students on the content of the four weeks of the electronic reading course by field experts. "Raz-Kids" was the platform used for reading online. Access to the program was granted by the GLOBED Company for the study to the sixth graders in the school. The program was initiated in 2004 and it is an internet-based computer program for young readers around the world (http://www.raz-kids.com). The program's features let students listen to fluent reading, record their own practice reading, and take a quiz at the end of reading. It also encourages teachers to set up individual accounts for each student so that an individual student can engage on their own level in independent reading. Points are given as a reward to listen to and read the story and take the quiz. Such points can be used to buy items that are included in the software as a game option for a learner's Raz Rocket. When leveling their books, the software used qualitative and quantitative data. It used a system of letters to describe each level of the book. The level of the book was associated with other common leveling programs like Developmental Reading Assessment, Lexile, and Reading Recovery.

The features of the gamified e-book platform are progress bar, leaderboard, badges, customizing the avatar/environment, fixed reward schedule, and immediate feedback. Participants used the platform at school in their EFL lessons.

SURVEY OF ADOLESCENT READING ATTITUDES

The Adolescent Reading Attitudes Survey (ARAS), developed by McKenna et al., (1995) determines the reading attitudes of adolescents regarding recreational and academic content in either print or digital format, and is scored to reveal positive, somewhat positive, neutral/indifferent, somewhat negative or negative attitude about recreational reading in print settings, recreational reading in digital settings, academic reading in print settings, and academic reading in digital settings. Bastug and Keskin (2013) adapted the questionnaire to the Turkish setting and permission to use the questionnaire was received

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from them. Bastug and Keskin (2013) tested the adaptation of the Adolescent Reading Attitudes Survey on 702 sixth, seventh, and eighth-grade students in Konya - 343 of them being female and the remaining 359 being male. The exploratory factor analysis and confirmatory factor analysis revealed that the four original sub-dimensions were retained when two items were omitted. Cronbach's alpha coefficients were 0.691, 0.802, 0.690, 0.606, and 0.623 for the overall scale, RD, RP, AP, and AD respectively, suggesting that the adapted scale was reliable and valid.

One of the preferred methods within the scope of the reliability of the scale was to examine internal reliability. Cronbach-Alpha values were examined to examine internal reliability. The data obtained from the pilot study were analyzed descriptively via SPSS 22 and the alpha value of the related Cronbach was shown to be .94, which indicated that the test was statistically sufficiently accurate.

STRUCTURED INTERVIEWS

Structured interviews were conducted with the teachers in the chosen secondary school. The participants were asked four questions in the interviews, which were applied as a method of data collection to gather qualitative data. These questions were included in the interviews aimed at assessing the views of teachers on the positive/negative effects of the strategies used in gamified reading with regard to learning, peer engagement, learner-teacher interaction, and problems encountered during implementation and resolution of these problems and provided feedback. The researcher and two field experts in the field coded the data separately to supply interrater reliability. Miles and Huberman's (1994) formula which requires 80% agreement between the coders was used, and it indicated interrater reliability (P > 92%).

Instruction Procedure

Treatment Group 1: Gamified Linear Electronic Reading Program

The gamified linear electronic reading program (Figure 1) was implemented for 4 weeks with 40-minute sessions each week. All 24 students were required to log in to the program during each session. Each student had their own username and password, and the software was set to their level of reading. The timer was set at 40 minutes for students to listen, read, and take quizzes on the books identified for their level. During these lessons, students were not allowed to control other features of the program and students were given training and instructions by the researcher to develop their reading fluency and comprehension skills before the experiment started for guided reading. The researcher decided on the order of the books to be read by the learners.

Treatment Group 2: Gamified Non-linear Electronic Reading Program

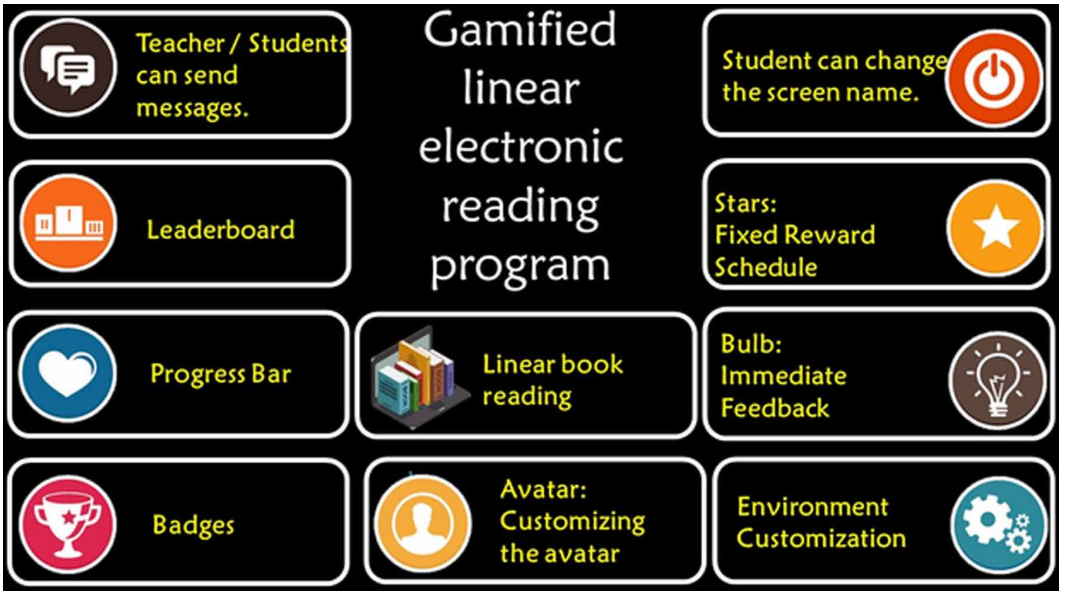
The gamified non-linear electronic reading program (Figure 2) was also implemented for 4 weeks with 40-minute sessions each week. All 24 students were required to log in to the program during each session. Each student had their own username and password, and the software was set to their level of reading. The timer was set at 40 minutes for students to listen, read, and take quizzes on the books identified for their level. During these lessons, students were not allowed to control other features of the program and students were given training and instructions by the researcher to develop their reading fluency and comprehension skills before the experiment started for guided reading. The researcher decided on the order of the books to be read by the learners.

Control Group: Printed Guided Book Reading Program

The printed guided book reading program (Figure 3) was presented for 4 weeks in each class with 40-minute sessions. All 24 students were required to read printed books during each session. Each child had their own printed book and the books appropriate for their level of reading were chosen. The timer was set at 40 minutes for students to read and take quizzes on the books identified for

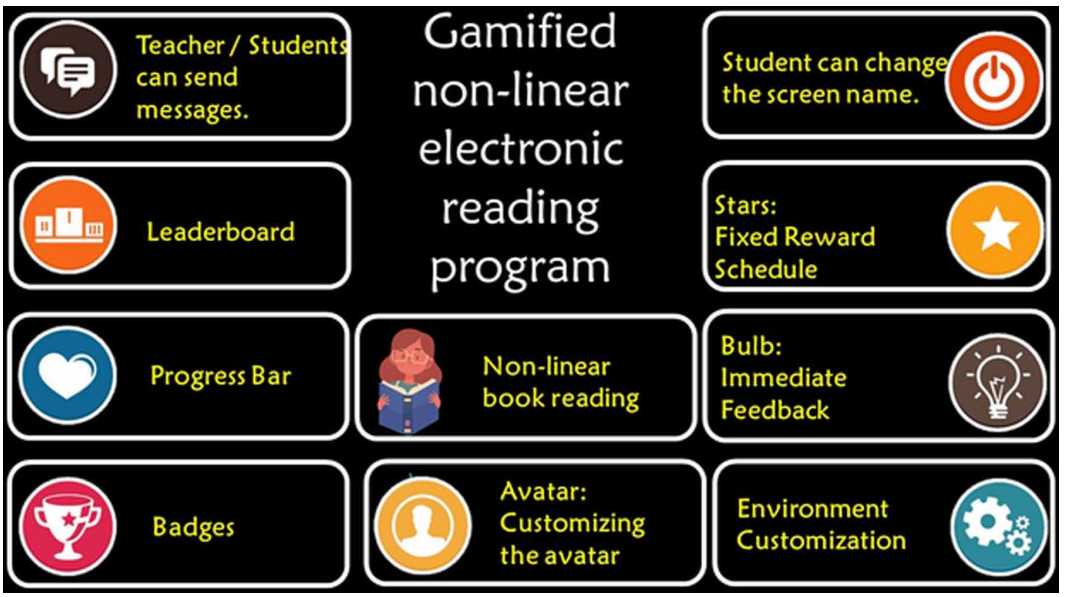
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Figure 1. Features of the Gamified Linear Electronic Reading Program



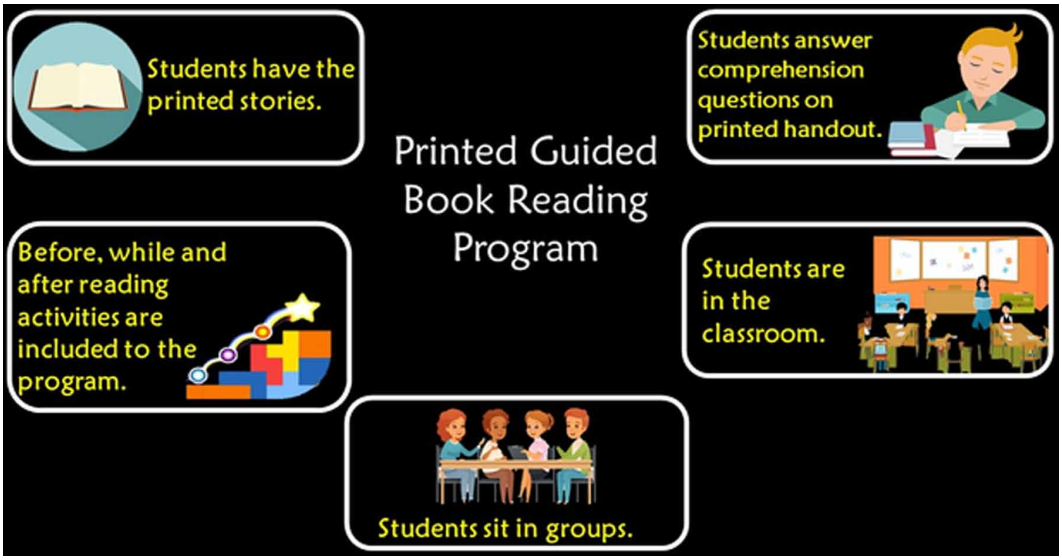
their level. During these lessons, the teacher acted as a guide for them and were given training and instructions by the researcher to develop their reading fluency and comprehension skills before the experiment started for guided reading. The researcher decided on the order of the books to be read by the learners.

Figure 2. Features of Features of the Gamified Non-linear Electronic Book Reading Program



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Figure 3. Features of the Printed Guided Reading Program



RESULTS AND DISCUSSION

Findings of the Influence of Linear Gamified Reading, Non-Linear Gamified Reading and Printed Book Reading on Reading Comprehension Levels

Data analysis showed a significant difference between the reading performance of EFL experimental and control groups in terms of their prose comprehension in pre- and post-tests. The comparison between these groups is shown in the Table 2. The table makes the comparison by using the mean score and standard deviation of the three groups.

The analysis shown in Table 2 suggests similar findings with previous research. The findings of the current study show that digital reading is effective in improving reading comprehension of the literary texts by EFL readers as the mean score of post-tests after the treatment of exposure to digital reading is higher than the pre-test scores in all cases.

Findings of the Influence of Gamified Linear Reading, a Non-Linear Gamified Reading and Printed Book Reading on Reading Attitude Levels

This analysis shows the overall e-book reading attitude of the experimental and control groups. A comparative analysis was made and the between-group statistics were tabulated to discover whether there was a statistically significant difference between the e-book reading attitude scores of the

Table 2. Comparison of Pre- and Post-Test Reading Comprehension of EFL Experimental and Control Groups

Groups		Mean	SD	F-value	p-value	Effect size
Gamified Non-linear Electronic Reading Group (n=24)	Pre-test Post-test	46.02 51.91	19.17 20.01	44.63 47.82	.000 .000	0.95 0.97
Gamified Non-linear Electronic Reading Group (n=24)	Pre-test Post-test	46.42 53.29	19.67 21.72	42.86 48.41	.000 .000	0.95 0.98
Printed Guided Book Reading Group (n=24)	Pre-test Post-test	46.22 51.84	19.56 20.02	44.85 47.76	.000 .000	0.95 0.97

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Table 3. Examination of Reading Attitude Measurement Invariance at Groups

Groups	Gamified Linear Electronic Reading Group (n=24)		Gamified Non-linear Electronic Reading Group (n=24)		Printed Guided Book Reading Group (n=24)		KW	P
	Mean	Sd	Mean	Sd	Mean	Sd		
Pre Academic Digital	3,95	0,76	4,85	0,32	3,45	0,79	13,83	0,00*
Post Academic Digital	5,18	0,54	5,70	0,38	3,60	0,79	29,97	0,00*
Pre Recreative Digital	2,76	0,83	3,15	0,52	2,04	0,26	37,60	0,00*
Post Recreative Digital	2,80	0,88	3,68	0,28	2,38	0,42	49,08	0,00*
Pre Academic Paper	2,26	0,70	2,73	0,64	1,66	0,00	30,06	0,00*
Post Academic Paper	2,25	0,69	2,93	0,45	1,73	0,13	40,49	0,00*
Pre Recreative Paper	3,29	2,03	4,33	2,12	2,45	2,02	8,75	0,03*
Post Recreative Paper	4,79	1,14	5,29	1,16	2,45	2,02	26,98	0,00*
Pre General E-book reading Attitude	4,22	1,13	4,43	1,06	3,64	1,25	9,21	0,02*
Post General E-book reading Attitude	4,75	0,60	4,87	0,71	4,08	1,01	9,77	0,02*

treatment and control groups due to the different reading environments: linear gamified reading, non-linear gamified reading, and printed book reading.

Table 3 shows that there was a statistically significant difference between the groups with regard to the pre academic digital scores and post academic digital scores in the e-book digital reading attitude questionnaire ($\chi^2(3) = 13,83,\ P = 0,00 < 0.05$). It was observed that pre academic digital scores of the non-linear gamified e-book reading group ($x\boxtimes = 4,85$) were higher compared to the scores of linear gamified e-book reading groups ($x\boxtimes = 3,08$). And a comparison of the pre academic digital scores of non-linear gamified e-book reading groups ($x\boxtimes = 4,85$) and printed book reading groups ($x\boxtimes = 3,45$) indicated the former to be higher than the latter.

Post academic digital scores of the students also differed significantly between the groups ($\chi^2(3) = 29,97,\ P = 0,00 < 0.05$). Nonlinear gamified e-book reading group's post academic digital scores ($x\boxtimes=5,70$) were higher than linear gamified e-book reading group's post academic digital scores ($x\boxtimes=4,89$). The non-linear gamified e-book reading group's post academic digital scores ($x\boxtimes = 5,70$) were also higher than printed book reading group's post academic digital scores ($x\boxtimes = 5,18$). A comparison of the post academic digital scores of linear gamified e-book reading group ($x\boxtimes = 5,70$) and printed book reading group ($x\boxtimes = 3,45$) also revealed the linear gamified e-book reading group's scores to be higher.

Post recreative digital scores of the students also showed a significant difference among the groups ($\chi^2(3)= 49,08,\ P = 0,00 < 0.05$). Non-linear gamified e-book reading group's post recreative digital scores ($x\boxtimes = 3,55$) were seen to be higher than printed book reading group's post recreative digital

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scores *(x⊠=2,80).* And linear gamified e-book reading group's post recreative digital scores *(x⊠=3,55)* were higher than printed book reading group's post recreative digital scores *(x⊠=2,38).*

Post academic paper scores of the students differed significantly among the groups as well *(χ^2(3)=40,49, P = 0,00 < 0.05).* It was seen that non-linear gamified e-book reading group's post academic paper scores *(x⊠=2,69)* were higher than printed book reading group's post academic paper scores *(x⊠=1,73).* Also, the linear gamified e-book reading group's post academic paper scores *(x⊠=2,93),* were higher than printed book reading group's post academic paper scores *(x⊠=2,25).*

There was a significant difference between the post recreative paper scores of the different groups *(χ^2(3)=26,98, P = 0,0 < 0.05).* When compared to the post recreative paper scores of the printed book reading group *(x⊠=2,45),* the non-linear gamified e-book reading group's post recreative paper scores *(x⊠=5,29)* were seen to be higher. The linear gamified e-book reading group's post recreative paper scores *(x⊠=4,79)* were also higher than the printed book reading group's post recreative paper scores *(x⊠=2,45).*

Finally, the comparison of the groups in terms of their post general reading attitude scores also revealed a significant difference between the groups *(χ^2(3)=9,77, P = 0,00 < 0.05).* The non-linear gamified e-book reading group's post general reading attitude scores *(x⊠=4,87)* were higher than the linear gamified e-book reading group's post general reading attitude scores *(x⊠=4,75).* The non-linear gamified e-book reading group's post general reading attitude scores *(x⊠=4,87),* were also higher than the printed book reading group's post general reading attitude scores *(x⊠=4,08).*

Focus Group Interviews with Participant Students after the Experiment

The interviews were transcribed from the recordings. The analysis and subsequent coding of transcripts revealed some common themes and the findings below are presented under these themes. In order to support the content of the themes, direct quotations of the participants were included.

The qualitative interviews with the respondents after the post-test revealed that most of the EFL students preferred to read on their screens digitally rather than using old-fashioned print reading. In total, 83% of EFL sixth grade students participating in the study showed a preference for digital reading rather than print reading. This high percentage confirms that today's generation, no matter what their native language is, tend to use modern technologies and be up-to-date all the time rather than being immersed in the traditional method of teaching and learning by the use of paper. In the focus group interviews, the students mentioned the reasons for their choice as being able to search online quickly and with no difficulty if they have difficulty in understanding any word or even the whole text, being equipped with audio or visual elements in the written text, or even the exciting pleasure of using the technology itself. However, the little percentage of the rest of the respondents who indicated a preference for print reading mentioned that they can understand better if they have a pen/pencil in their hand and take some notes, underline, circle, or highlight any parts that they think they need to pay more attention to. Some of them also mentioned that they are able to read and understand uisng both methods but because for years they were trained to use print texts, they just prefer to have the text physically on paper.

Learner Autonomy

The first theme emerging from the interview data was about how electronic reading influenced students' EFL reading process and habits. When the transcripts of focus group interviews were analyzed, it was possible to state that students who were reading from the screen thought they were responsible for their own learning and many students claimed that they found electronic reading quite enjoyable and they considered electronic reading as a comfortable environment to read stories compared to a guided printed book reading group. It was also noticed that reading texts online helped them feel more confident. Below are some excerpts from student interviews regarding these findings:

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[...] I felt in a rush when I read the story in the lesson. With the help of time, I think I got used to it. When I read the stories I felt responsible and motivated to read the stories. (S2, Non-linear Gamified E-book Reading Group, Interview Data, 05.11.2019)
[...] Everyone in the online class was reading stories in the lesson and we have never done this before which made the lesson more interesting. Each story's ending and main characters were different. We compared the stories in the break time. The best thing was to get the help of the teacher. (S5, Linear Gamified E-book Reading Group, Interview Data, 05.11.2019)

These excerpts illustrate that e-book reading may have encouraged students to improve their *"self-regulated learning skills"* since they have recognized that they were able to learn at their own pace. They also stated they were more actively engaged in the stories and most of the students talked about the confidence and motivation they acquired.

Online or Printed

In the interviews, the students were also asked whether they preferred electronic or printed material to read at school. Nearly all of the students stated that they would like to read online at school, but they also mentioned that they understand better when the text is printed.

Below are some excerpts from the answers of the students:

[...] I love when I buy printed books (S5, Linear Gamified E-book Reading Group, Interview Data, 05.11.2019)
[...] Storybooks are my favorite to read. I buy printed books. Until this time, I have never bought electronic books (S5, Non-linear Gamified E-book Reading Group, Interview Data, 04.11.2019)
[...] I underline the important parts in school course books (S2, Non-linear Gamified E-book Reading Group, Interview Data, 05.11.2019)

Although students were more motivated to read e-books, they claimed they preferred to read printed books in their free time as a recreational activity. They also mentioned that they remember more when they read printed text.

Immediate Feedback

In the focus group interviews, students in the non-linear gamified and linear gamified e-reading groups were asked questions about the advantages of using electronic reading. Nearly all the students said that they would like to read online at school as they get immediate feedback.

Some excerpts from student replies:

[...] It was good to take feedback, to see the answers after we completed. (S4, Non-linear Gamified E-book Reading Group, Interview Data, 04.11.2019)
[...] When we finish the reading, the rocket flew. It was cool. (S1, Linear Gamified E-book Reading Group, Interview Data, 04.11.2019)

Students in the printed reading group were also asked about the disadvantages of using the reading program. Nearly all of the students stated that they would like to get immediate feedback.

Some of the answers of the participants:

[...] We answered the test but the teacher collected those papers and gave feedback on the other week after she checked the results. (S2, Printed Book Reading Focus Group, Interview Data, 05.11.2019
[...]We did not learn the correct answers weekly. (S1, Printed Book Reading Focus Group, Interview Data, 05.11.2019)

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Participants of the printed book reading sessions complained about not getting immediate feedback from their teacher. They also noted that as the answer key was given verbally, they could not take the notes properly, which suggests an advantage for the electronic books as they are more feasible in terms of students getting immediate feedback and being able to see the answers individually on the screen rather than hearing them.

Focus Group Interviews with Participant Teachers after the Experiment

The findings showed that the observer teacher of the non-linear gamified and linear gamified e-book reading groups felt satisfied with the lesson in different ways such as the learning environment and the engagement it creates. And teachers also commented on some of its negative aspects.

Learning Environment

The participants were found to be actively practicing independent reading during the e-book reading sessions. The replies from observing teachers suggested that this learning environment might increase reading comprehension and autonomy of the learners.

The statements below illustrate these findings:

[…] Class read online and I observed them they look really engaged. When they were reading their cameras were on, when they do not understand they asked questions. (T1, Non-linear Gamified E-book Reading Group, Journal Data, 23.10.2019)
[…] When they come across an unknown vocabulary, they use an online dictionary and note the new word in their online notebooks. They learned new vocabulary alone, while they were reading, doing some internet research. (T2, Linear Gamified E-book Reading Group, Journal Data, 23.10.2019)

All in all, teachers' reflections reveal that e-book reading lessons offered students the ability to plan their path of learning or discover new ways of learning on their own. In addition, it could also be concluded that such an "autonomous learning environment" provided students with the ability to learn in individually diverse ways. They also added that even though all the participants had gone through some pedagogical and technical challenges, they were willing to use electronic reading; and their answers indicated that this was mainly because they believed the necessity to make use of technology in teaching and had already witnessed its benefits especially in terms of motivating the students.

Ownership and Reader Preferences

When analyzing the results, it was seen that teachers in all groups mentioned that they prefer owning and reading print books over e-books.

Some of their comments are extracted below:

[…] I do not think that reading from screen is useful. When I read something, I need to take notes, and color the parts which are important. When I read online, I do not have this chance. (T1, Linear Gamified E-book Reading Group, Journal Data, 10.10.2019).
[…] It does not give me the pleasure of printed text. (T2, Linear Gamified E-book Reading Group, Journal Data, 10.10.2019).
[…] When I have the printed book in my hand, I like the texture of it, I love turning pages or underlining the sentences I like. (T4, Printed E-book Reading Group, Journal Data, 10.10.2019).

Overall, participant teachers seem to favor printed books as it gives them the chance to underline, and to take notes. Owning the printed book, underlining the important parts, and the texture of the book are still important habits that they cannot leave easily.

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DISCUSSION AND CONCLUSION

The current research explored the effects of electronic reading classes on reading comprehension, reading attitude and reflections of sixth grade Turkish EFL students and teachers.

In this study, the impacts of gamified reading comprehension activities on students' reading attitude were analyzed. The research findings showed that all students who participated the study had improved their reading skills. In addition, it was found that the students had positive attitudes towards electronic reading as they were able to accomplish their goals more easily at their respective L2 levels and alone. The results of this study indicated that there is a statistically significant difference in reading attitudes of learners exposed to different instructional reading environments.

The findings indicated that the participants in the printed book reading group had lower reading attitude scores than the ones in non-linear and linear gamified e-book reading groups. A strong relationship between reading attitude and technology integration in classes were mentioned in previous studies (Baker & White, 2010; Hamari & Eranti, 2011). Therefore, it is possible to state that the use of electronic reading has a vital place in EFL classes and when students use digital platforms to read, their attitude might change towards positive especially when reading for academic purposes. In this research, the attitudes of sixth grade students towards reading at a high level had been found to be positive which is in line with the findings of many previous studies (Balcı et al., 2012; Durualp et al., 2013; İşeri, 2010).

The results showed that the non-linear gamified e-book reading group's academic digital scores were the highest. The reason behind this could be that students consider reading storybooks in class time as academic reading. When post recreative digital and paper scores were examined, it was found that students prefer to read printed books over e-books as a leisure activity, which calls for more electronic reading opportunities to change this impression. Another study shows that exposure to books from sources other than the library often predicts achievement in reading (Evans et al., 2010). As schools transition to paperless classrooms across the globe, the use of digital devices as reading resources has become increasingly relevant (Giebelhausen, 2015; Shishkovskaya, Sokolova, & Chernaya, 2015). Paperless classrooms enable the reader to adjust the text size, highlight important passages and search for relevant words outside of the text by clicking a button.

Overall, our results corroborate those of previous research, regarding differences in reading attitudes according to gender and type of reading (leisure vs. academic), showing a persistent gender effect among pupils in grade 4, favoring girls (McKenna et al., 1995, 2012), and more positive attitudes toward leisure reading than toward academic reading, even though there was a negative trend in attitudes toward both types of reading across grades. However, our finding that there are significant differences in reading attitudes were restricted to the period of sixth grade. The how and why of these changes during sixth grade in our study context are still unclear, but the transition should probably be regarded as one of the factors influencing young people's attitudes toward reading, at least in some contexts. These discrepant results also suggest that we cannot assume that the results of research on this subject are transferable, thus underlining the relevance of closely documenting these differences and changes in attitudes toward reading among pupils in the course of their schooling in a range of linguistic and education contexts.

The novelty of this research in the field of reading comprehension in digital environments lays in using gamified reading as the focus rather than e-books. This was also one of the few studies that examined storybooks, not educational books, in the area of gamification. The analysis contributed either to reading comprehension or gamified storybook reading, and it calls experts to consider conducting new research on the reading comprehension performance of the students using educational gamified books rather than the storybooks.

This study not only analyzed the perceptions of the students, but also teachers about gamified e-reading. The students in the study stated that they liked the tactile feature of the printed books. However, as young readers still tend to depend on the suggestions of the adults who are close to them

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(Zasacka, 2017) the researcher contends that teachers, parents and eventually the students should be made more aware and informed of possible e-learning platforms or online electronic library sources. In Poland, Zasacka (2017) carried a comparative study on 12-13 year old learners to understand their reading behavior and digital media habits. In that study, 70% of the participants reported that they read texts which are created by their peers (blogs and fanzines). 37% percent of the participants revealed that they actively contribute content by creating their own websites and blogging (Zasacka, 2017). Nine percent of the respondents stated that they read books as a leisure activity (Zasacka, 2017).

As the current study indicates, the power of technology in reading classes lies in the instant and formative feedback it provides. Both students and their teachers in the experimental groups talked about the instant and formative feedback as an advantage online learning platforms supply but it should be noted that online reading platforms alone would not be sufficient to take advantage of formative assessment in the classroom. Along with it, the culture of the class should change, the atmosphere of trust and student centeredness should be the focus. When what we say and what we do contradicts, students learn what we do. According to Demir (2017), all types of immediate feedback interventions have a positive influence on transfer of learning. The same author further adds that immediate feedback has a constructive influence on learning transfer. Feedback is a task that needs to be handled with care and precision. An analysis study that reviewed 131 studies on feedback reveals that more than a third of feedback interventions actually reduce student performance (Kluger & DeNisi, 1996). Feedback that emphasizes especially negative aspects of the student performance and focuses only on correction can produce negative outcomes. Delayed feedback also loses its effect. Therefore, an effective, meaningful and understandable feedback should have the following characteristics: it should be on time, see effort, be specific, be process-oriented, organized, focus on development, ensure the transfer of knowledge, be mobilized, be tailor-made, cover the process, include more than just praise, and be appropriate for features and goals.

The largest social experiment in world history was conducted with about 1.6 billion students with the help of COVID-19 (Anderson, 2020; Zimmerman, 2020). When the COVID-19 pandemic began in Turkey, it was the middle of the spring term and as in many other countries all around the world, universities and schools had to turn quickly to online and interactive education platforms, embracing remote forms of teaching and learning when closures of the schools and universities were announced (Liman Kaban, 2020). Especially after COVID-19 the link between technology and learning has become inseparable. With this process, it is inevitable that these experiences will have reflections on the general education paradigm in the long run. Every crisis generates opportunities in its continuation. In this context, success as well as many failures stories and acquired experiences can be turned into an opportunity. Before COVID-9 struck the world not only the educators but also students were looking from afar against screen reading and were not familiar with online reading. With the assist of pandemic teaching, millions of students around the world had the chance to try electronic reading. It is possible to conclude that digital reading strategies will gain importance. Schools and administrators need to involve teachers in the design of policy. If they do not involve teachers to the process, teachers are not going to help you with implementation.

In summary, the results of the current study demonstrate that in EFL classes the use of screen reading lessons while teaching and learning reading can be considered an effective way to promote the growth of reading comprehension skills and can improve reading performance and reading attitude levels.

The findings of qualitative as well as quantitative data of this study prove that the digital reading of English reading comprehension texts affects the comprehension of such texts by EFL readers significantly in a positive sense. The successful performance of the experimental groups demonstrates that the digital reading of reading comprehension texts improves the EFL readers' understanding due to comprehensible input. Moreover, it was found that both groups of EFL readers prefer to read digitally rather than on print. A pedagogical implication that can be drawn from this study is that the language and literature teachers and course developers can provide more interactive and convenient

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learning environment for the students with the help of digital resources that can motivate the students to learn in a way that they are more interested in and have more enjoyable experiences in their learning process. Although the significance of digital reading comprehension of English texts is confirmed in this study in both EFL contexts, it is still a long way for this method to be used in practice around the world.

Conflicts of Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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MISSION

The primary mission of the **International Journal of Mobile and Blended Learning (IJMBL)** is to provide insight and understanding into the role of innovative learning theory and practice in an increasingly mobile and pervasive technological environment. As technology enables a more seamless experience of device-supported learning worlds that may integrate mobile, embedded, augmented, and immersive technologies, researchers, professionals, and academicians may expect to see increasing interest and activity in blended approaches to learning. IJMBL brings together experts at the forefront of this field, in both technology and pedagogical practice, and assists them in the development and dissemination of new approaches to both mobile and blended learning.

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